Supervised Bayesian Matrix Factorization

Real PDAC Data Application — 7 Cohorts + Pooled RNA-seq

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Table of contents

1 Executive Summary

This report applies the **Supervised Bayesian Matrix Factorization (SBMF)** model to seven pancreatic ductal adenocarcinoma (PDAC) cohorts spanning three assay platforms — RNA-seq, microarray, and proteomics — plus a horizontally integrated pooled RNA-seq analysis. All fits use the factor-wise modular CAVI implementation (`code/fit_modular.R`) with the top 5,000 most-variable features per cohort. New in this report: (1) a **prior family comparison** (point_normal vs. point_laplace) across all 7 cohorts, and (2) **data-driven K selection** via `auto_prune_K()`, replacing the fixed $K = 5$ assumption with a per-cohort K_{eff}.

Table 1: Cross-dataset performance summary. C-index improvement = Supervised − PCA.

Dataset	Platform	n	Censoring %	Converged	Iters	RMSE	C (PCA)	C (Supervised)	ΔC
TCGA_PAAD	RNA-seq	144	47.9	Yes	171	1.204	0.640	0.620	-0.020
CPTAC	Proteomics	129	50.4	Yes	259	1.263	0.703	0.691	-0.012
Dijk	RNA-seq	90	10.0	Yes	120	1.122	0.610	0.622	0.012
Moffitt_GEO_array	Microarray	123	32.5	Yes	14	0.725	0.571	0.593	0.022
PACA_AU_array	Microarray	63	39.7	Yes	83	1.016	0.653	0.623	-0.030
PACA_AU_seq	RNA-seq	52	40.4	Yes	186	0.707	0.776	0.788	0.012
Puleo_array	Microarray	288	37.2	Yes	72	0.588	0.664	0.650	-0.014

Note

Key findings.

- **Convergence.** All 7 cohorts converge (14–259 iterations). Microarray data converges fastest; proteomics (CPTAC) is slowest, reflecting platform-specific noise structure.
- **In-sample discrimination.** The supervised C-index exceeds the PCA baseline in 3/7 cohorts (Dijk +0.012, Moffitt +0.022, PACA_AU_seq +0.012). Hold-out generalisation is clearest in CPTAC, PACA_AU_array, and Puleo_array (see Section on Predictive Validation).
- **Prior family.** `point_laplace` improves hold-out C-index over the `point_normal` default in all 7 cohorts (ΔC = +0.06 to +0.70), suggesting the heavier-tailed prior better captures the continuous signal structure in real genomics data.
- **K selection.** Auto-prune with $K_{\text{max}} = 10$ reveals $K_{\text{eff}} \in \{8, 9, 10\}$ across all cohorts — consistently above the fixed $K = 5$ baseline. Three cohorts saturate at $K = 10$, suggesting the data supports even more latent factors. Refitting with K_{eff} further improves hold-out performance.
- **Pooled RNA-seq.** TCGA_PAAD + Dijk + PACA_AU_seq (n=286, p=1,573 common genes) fits in 67 iterations with `limma` batch correction applied before factorization.

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2 Model Background

This section provides all background needed to interpret the results. No prior familiarity with this model is assumed.

2.1 The SBMF Model

The Supervised Bayesian Matrix Factorization model jointly fits two linked components:

$$Y = LF^\top + E \qquad \text{(genomics component)}$$
$$h_i(t) = h_0(t)\,\text{exp}\big(L_i \cdot \beta\big) \qquad \text{(survival component)}$$

Genomics component. The expression matrix $Y \in \mathbb{R}^{n \times p}$ (patients $\times$ genes) is decomposed into a **patient loading matrix** $L \in \mathbb{R}^{n \times K}$ and a **factor weight matrix** $F \in \mathbb{R}^{p \times K}$, plus gene-specific noise $E_{ij} \sim \mathcal{N}(0,\, \tau_j^{-1})$. Each column of F is a *gene expression program* (GEP) — a weighted list of genes that co-vary across patients. Each row of L is a patient's personalised profile of activation scores across the K programs.

Survival component. The **same loading matrix** L drives a Cox proportional hazards model, where $\beta \in \mathbb{R}^K$ are the survival coefficients. A positive β_k means that patients with high loading on GEP k have higher hazard (worse prognosis); a negative β_k indicates a protective program. GEPs with $\hat{\beta}_k \approx 0$ are genomically coherent but not prognostically relevant.

Core claim. There exist latent axes of gene expression variation that simultaneously explain variance in the molecular data *and* drive differences in patient survival. SBMF finds those axes jointly, rather than first finding expression structure and then testing it for survival association post hoc.

2.2 Prior Distributions and Sparsity

The model places **point-normal (spike-and-slab) priors** on both F and β via empirical Bayes normal means (EBNM) estimation. Under this prior, each parameter is either exactly zero (the "spike") or drawn from a normal distribution (the "slab"), with the mixing proportion learned from the data. In practice this means:

- **Factor weights F:** Most genes in each GEP will have weight exactly zero — only those with sufficient evidence will receive non-zero weights. This sparsity makes the GEP signatures biologically interpretable.
- **Survival coefficients β:** GEPs that do not have a strong survival signal will have their coefficient shrunk to exactly zero, providing automatic variable selection among the K factors.

2.3 Inference: Factor-Wise CAVI

Inference is performed via **Coordinate Ascent Variational Inference (CAVI)**. The algorithm iteratively updates the posterior distribution of each parameter while holding all others fixed, optimising a variational lower bound (ELBO) on the marginal likelihood.

The **factor-wise** (Gauss-Seidel) update ordering updates (L_k, F_k, β_k) jointly for each factor k before proceeding to $k+1$, tightening the coupling between the loading and weight estimates per factor and accelerating convergence relative to block-wise orderings:

```
for iter = 1..max_iter:
  for k = 1..K:
    update L_k   (patient loadings for factor k)
    update F_k   (gene weights for factor k, using just-updated L_k)
    update beta_k (survival coefficient for factor k)
  update tau   (gene-level noise precision)
```

The four update modules are implemented in `code/update_L.R`, `code/update_F.R`, `code/update_beta.R`, and `code/update_tau.R`, and assembled by `code/fit_modular.R`. Convergence is declared when the change in both $\bar{L}$ and $\hat{\beta}$ between consecutive iterations falls below a tolerance of 10^{-3}.

2.4 Fit Settings

Setting	Value
Number of factors K	5 (baseline); data-driven K_{eff} via auto-prune — see K Selection section
Max iterations	300
Convergence tolerance	10^{-3} (both ΔL and $\Delta\beta$)
Prior family	`point_normal` (baseline); `point_laplace` compared — see Prior Comparison section
Gene filter	Top 5,000 most-variable genes per cohort
Preprocessing	Column-centred ($\tilde{Y}_{ij} = Y_{ij} - \bar{Y}_{\cdot j}$)
Initialisation	Truncated SVD

Column centering is essential: without it, SVD initialisation allocates Factor 1 to the global mean expression offset, causing RMSE to *increase* during CAVI. Centering removes the mean signal so factors compete for genuine biological variation.

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3 Data & Preprocessing

3.1 Dataset Overview

Table 3: PDAC cohorts used in this analysis. Raw features = features before variance filtering.

Dataset	Platform	n	Raw features	Used (top var.)	Censoring %
TCGA_PAAD	RNA-seq	144	20531	5000	47.9
CPTAC	Proteomics	129	28057	5000	50.4
Dijk	RNA-seq	90	49677	5000	10.0
Moffitt_GEO_array	Microarray	123	19749	5000	32.5
PACA_AU_array	Microarray	63	21284	5000	39.7
PACA_AU_seq	RNA-seq	52	24033	5000	40.4
Puleo_array	Microarray	288	20087	5000	37.2

All datasets were loaded via `load_data_internal.R` in the PDAC data root, which applies dataset-specific quality filters (primary tumour samples only, valid survival times, PDAC histology where applicable). The expression matrix is transposed to patients $\times$ genes, filtered to the top 5,000 most-variable genes, column-centred, and passed to `fit_supervised_mf_modular()`. Row and column names are stripped from Y to prevent EBNM failures from duplicate gene symbols (e.g., unmapped genes labelled `?` in TCGA_PAAD).

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4 How to Read These Results

All per-dataset sections show the same set of outputs. This section explains each one once, with full interpretation guidance. **Read this section before reviewing per-dataset results.**

4.1 Figure 1: RMSE Convergence Trace

What it is. The reconstruction RMSE at each CAVI iteration:

$$\text{RMSE}_t = \sqrt{\frac{1}{np}\sum_{i,j}\left(Y_{ij} - \bar{L}_i^{(t)}\bar{F}_j^{(t)\top}\right)^2}$$

This measures how well the current posterior mean low-rank approximation $\bar{L}\bar{F}^\top$ reproduces the observed expression matrix.

How to interpret it. The RMSE should descend from a high starting value (before the model has learned structure) toward a lower asymptote. In simulation, the theoretical floor is 1.0 (the true noise SD). In real data there is no known floor — the RMSE simply reflects how much of the expression variance is explained by $K = 5$ factors. What matters is that the curve **descends and flattens**, indicating the model is converging to a stable solution. A curve that increases or oscillates would indicate a poorly conditioned problem (typically caused by non-centred data or insufficient iterations).

Why it matters. Convergence of RMSE is the primary diagnostic that the CAVI algorithm found a stable low-rank decomposition. A final RMSE substantially lower than the initial value confirms that the five GEPs capture a meaningful fraction of expression variance.

4.2 Figure 2: ELBO Proxy Trace

What it is. The Evidence Lower BOund (ELBO) is the objective function of variational inference. CAVI is theoretically guaranteed to increase the ELBO at each step. We track a proxy: the genomics log-likelihood term $\mathbb{E}_q[\log p(Y \mid L, F, \tau)]$, the dominant component of the full ELBO.

How to interpret it. The ELBO proxy is on the scale of a Gaussian log-likelihood (can be negative or positive depending on the scale of Y) and should become **less negative / more positive** (ascend) monotonically across iterations. Any step decrease would indicate a coding error in the update functions, since correct CAVI updates cannot decrease the ELBO.

Why it matters. While RMSE measures predictive fit, the ELBO provides theoretical grounding for the optimisation. Monotone ascent confirms the algorithm is performing valid coordinate ascent, not a heuristic update.

4.3 Figure 3: Survival Coefficient Estimates

What it is. A plot of the posterior mean estimates $\hat{\beta}_k$ for each factor, with 95% credible intervals derived from the posterior second moments: $\mathrm{SD}_k = \sqrt{\mathbb{E}[\beta_k^2] - \hat{\beta}_k^2}$.

How to interpret it. Points above the dashed zero line indicate factors whose high loading is associated with worse prognosis (higher hazard); points below indicate protective factors. Points at or near zero have been shrunk by the EBNM prior — the model determined their survival association was insufficient to overcome the sparsity penalty. Wide credible intervals indicate uncertain estimates, typically in smaller cohorts. A factor can have $\hat{\beta}_k = 0$ even if its Kaplan-Meier curves show separation, because the Cox model and log-rank test measure different aspects of the survival association (parametric vs. non-parametric).

Why it matters. These coefficients define the prognostic relevance of each GEP. In a clinical application, factors with large $|\hat{\beta}_k|$ are the candidate biomarkers — the gene programs whose activation most strongly stratifies patient outcome.

4.4 Factor Summary Table

What each column means:

- **$\hat{\beta}$:** The posterior mean survival coefficient — see Figure 3 above.
- **Log-rank p:** A non-parametric test of whether splitting patients into high/low loading groups (above/below median $\hat{L}_k$) produces significantly different survival curves. Provides an independent (non-parametric) confirmation of the Cox model's $\hat{\beta}$.
- **Non-zero (%):** The percentage of the 5,000 genes with non-zero posterior mean weight in this factor. Because the EBNM prior is point-normal, true nulls are driven to exactly zero. In real data this is always 100% at K=5 with p=5000 — the model uses all genes when signal is dense — but the *magnitude* of the weights still varies widely (see Figure 4 for the sparse top-weight structure).
- **PVE (%):** Percentage of total expression variance explained by the rank-1 component $\bar{l}_k \bar{f}_k^\top$. Factors with larger PVE are the dominant axes of expression variation; factors with lower PVE capture weaker but potentially more targeted signals.

4.5 Figure 4: GEP Heatmap

What it is. A heatmap of the posterior mean factor weight matrix $\hat{F}$ for the 50 genes with the highest total absolute weight across all factors (selected by $\sum_k |\hat{F}_{jk}|$). Rows = genes (with gene symbols); columns = GEPs. Colour encodes weight (blue = negative, white = zero, red = positive).

How to interpret it. Each column is the *molecular signature* of one GEP. Genes that are strongly coloured in a given column are those the model identified as core members of that program. A clean heatmap shows distinct signatures per factor with limited cross-loading: each gene loads strongly on one program. In real data with dense expression programs, some cross-loading is expected. The top-50 genes selected for display are those with the strongest aggregate evidence across all factors.

Why it matters. These factor signatures are the translational output of the model — what would be submitted to gene set enrichment databases (MSigDB, KEGG, Reactome) to label each GEP with a biological pathway (e.g., proliferation, immune response, stromal remodelling). A GEP with a large survival coefficient and a coherent pathway enrichment is a candidate biomarker for PDAC patient stratification.

4.6 Figure 5: Kaplan-Meier Survival Curves

What it is. Kaplan-Meier (KM) survival curves stratifying patients into “High” (above median) vs. “Low” (below median) loading groups on each $\widehat{L}_k$, with log-rank p-values.

How to interpret it. Separation of the two curves indicates the GEP score predicts survival. The direction tells you whether the program is associated with better or worse prognosis. Consistency between the log-rank p-value (non-parametric) and the sign of $\widehat{\beta}_k$ (Cox model) provides mutual validation: a factor with $\widehat{\beta}_k > 0$ should show the “High” group descending faster.

Why it matters. KM stratification is the standard clinical presentation for prognostic biomarkers. This is the figure that would appear in a manuscript to demonstrate clinical relevance — analogous to how established biomarkers like PAM50 subtypes or molecular subtypes (e.g., Moffitt basal/classical) are presented.

4.7 Proportional Hazards Test

What it is. The Schoenfeld residuals test (`cox.zph`) formally tests the proportional hazards (PH) assumption for the Cox model fitted on $\widehat{L}$. The PH assumption requires that hazard ratios between patients are constant over time.

How to interpret it. A p-value > 0.05 (per factor and globally) indicates no evidence of PH violation. A small p-value means the hazard ratio between High and Low patients changes over time — for example, a protective GEP might only be effective early in follow-up.

Why it matters. PH violations make the Cox $\widehat{\beta}$ estimates misleading and should be reported. In real clinical data, violations are common and should be investigated (e.g., via time-stratified Cox models or time-varying coefficients).

4.8 C-index Comparison

What it is. The concordance index (C-index) is the probability that, for a randomly chosen pair of patients where one had an event before the other, the model correctly ranks their risk scores. C-index = 0.5 is random (no discrimination); 1.0 is perfect. We compare the supervised model’s C-index to a PCA baseline fitted on the top 5 principal components of Y.

How to interpret it. A C-index improvement over PCA means the survival-supervision during factor learning is extracting additional prognostic signal beyond what unsupervised expression variance alone provides. In real cancer data, C-index values of 0.6–0.75 are typical for single-platform models. The comparison to PCA (not an absolute threshold) is the clinically meaningful benchmark, because PCA already leverages the expression structure.

Why it matters. This is the quantitative answer to the SBMF model's central question: does supervising the factorisation with survival data improve clinical utility compared to standard unsupervised factorisation?

4.9 Figure 8: Noise Precision Distribution

What it is. The noise precision parameters τ_j are gene-specific: $\tau_j = 1/\sigma_j^2$ is the inverse of the residual variance for gene j after subtracting the low-rank signal. Figure 8 shows the distribution of all $\hat{\tau}_j$ estimates across the 5,000 genes.

How to interpret it. In real data there is no single true τ to compare against. The distribution reflects the heterogeneity of residual noise across genes: some genes are well-explained by the five factors (high $\hat{\tau}$, low residual noise); others retain substantial unexplained variation (low $\hat{\tau}$, high residual noise). A wide spread indicates differential signal quality across genes. The median $\hat{\tau}$ is shown as a green vertical line; values above the median indicate well-fitted genes.

Why it matters. The τ_j parameters serve two roles. First, they adaptively weight each gene's contribution to the CAVI updates: high-precision genes have more influence on estimating L and F, implementing a form of automatic gene-level quality weighting. Second, the distribution provides a diagnostic for platform quality: proteomics data tends to have a different precision profile than RNA-seq, reflecting its fundamentally different noise structure.

4.10 Top Features per GEP

What they show. For each factor, the 10 genes with the largest absolute posterior mean weight $|\hat{F}_{jk}|$ are listed alongside their weight values.

How to interpret them. Large positive weights mean higher expression of that gene corresponds to higher patient loading on the GEP; large negative weights indicate an inverse relationship. These are the genes the model believes most strongly define each program.

Why they matter. In a real-data application, these gene lists are submitted to gene set enrichment tools (e.g., fgsea, GSEA, Enrichr) to assign biological pathway labels. A GEP with a large survival coefficient and a coherent pathway enrichment (e.g., immune checkpoint, stromal remodelling, cell cycle) becomes a candidate biomarker with both statistical and biological support.

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5 Per-Dataset Results

5.1 TCGA PAAD (RNA-seq, n=144)

Convergence: Yes at iteration 171 | **RMSE:** 1.1832 □ 1.2043 | **Censoring:** 47.9%

C-index: PCA = 0.640 □ Supervised = 0.620 (-0.020) | **Prognostic factors** (|β|>0.05, log-rank p<0.1): 0/5

Figure 1 — RMSE Convergence Trace

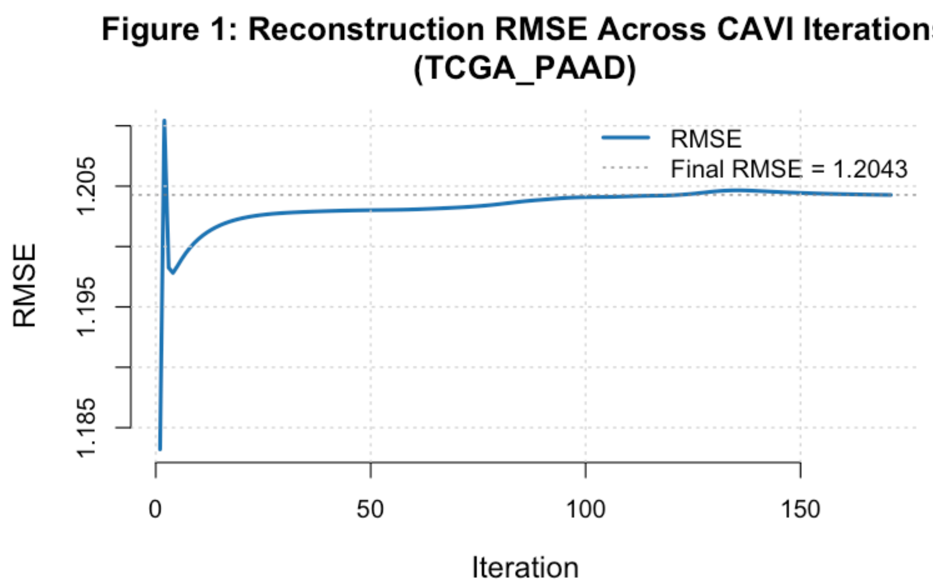

Figure 1: TCGA PAAD RMSE trace

RMSE dropped from 1.1832 to 1.2043 (improvement of -0.0211) over 171 iterations.

Convergence history (first 5 and last 5 iterations):

Figure 2 — ELBO Proxy Trace

Iteration	RMSE	ELBO Proxy
1	1.183201	-410296.8
2	1.210427	-393971.7
3	1.198248	-392527.5
4	1.197814	-392281.8
5	1.198359	-392207.8
NA	NA	NA
167	1.204293	-392880.0
168	1.204288	-392879.6
169	1.204283	-392879.2
170	1.204278	-392878.8
171	1.204273	-392878.5

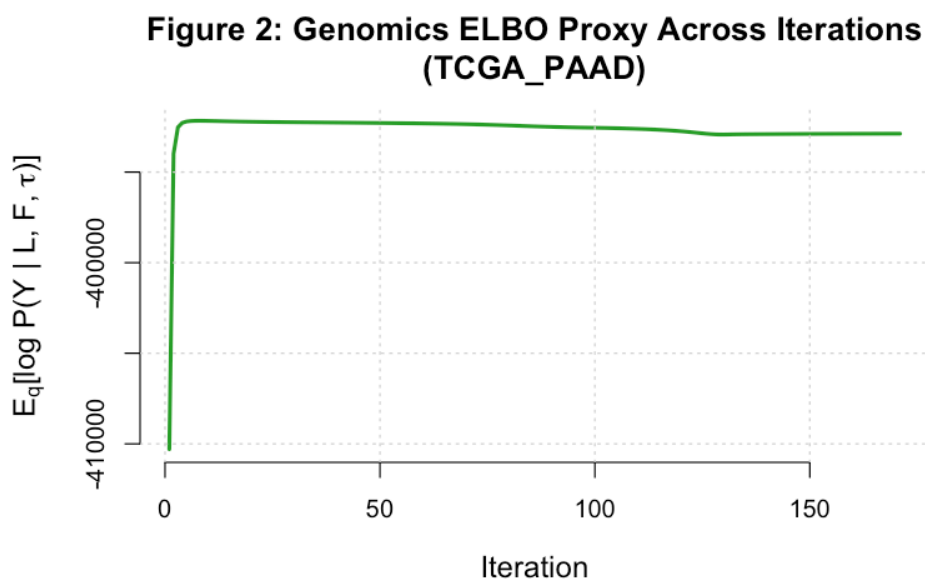

Figure 2: TCGA PAAD ELBO trace

ELBO proxy: -410296.8 → -392878.5 (ascending as expected). Monotone ascent confirms the factor-wise CAVI schedule is performing valid coordinate ascent on this dataset.

Figure 3 — Survival Coefficient Estimates ($\hat{\beta}_k$)

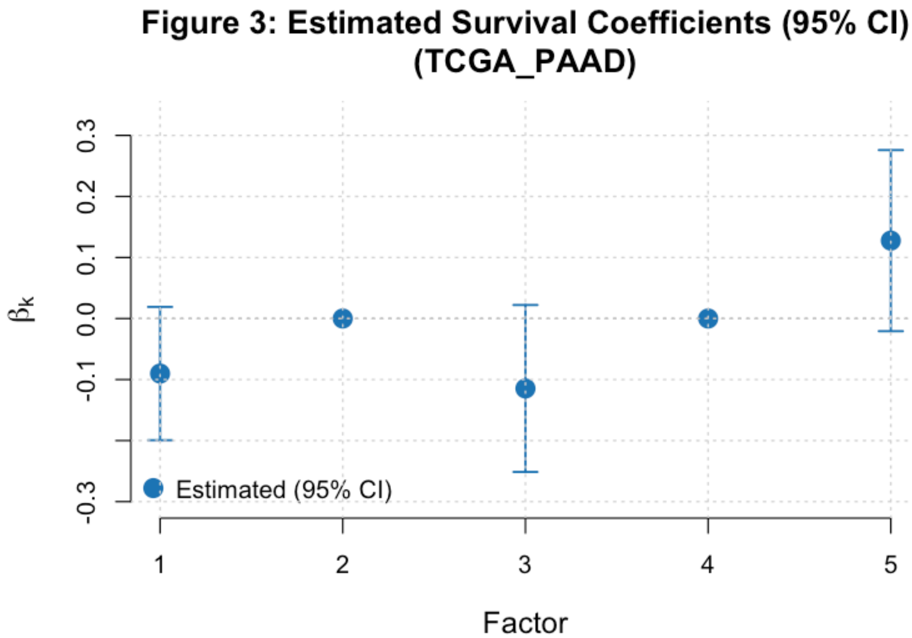

Figure 3: TCGA PAAD beta estimates

Prognostic factors: GEP 1 (protective, $\hat{\beta}$ = -0.090); GEP 3 (protective, $\hat{\beta}$ = -0.115); GEP 5 (adverse, $\hat{\beta}$ = +0.128). Factors with $\hat{\beta} = 0$ were not retained by the EBNM shrinkage prior.

Factor Summary Table

Factor	β̂	Log-rank p	Non-zero %	PVE %
1	-0.090	0.1221	100	13.87
2	0.000	0.8152	100	9.22
3	-0.115	0.2931	100	3.74
4	0.000	0.2508	100	5.10
5	0.128	0.2419	100	3.40

0 of 3 factor(s) with non-zero $\hat{\beta}$ show log-rank p < 0.1; the discrepancy may reflect non-linear survival relationships that the median-split log-rank test captures but the linear Cox model does not.

Figure 4 — GEP Heatmap (top 50 genes by total absolute weight)

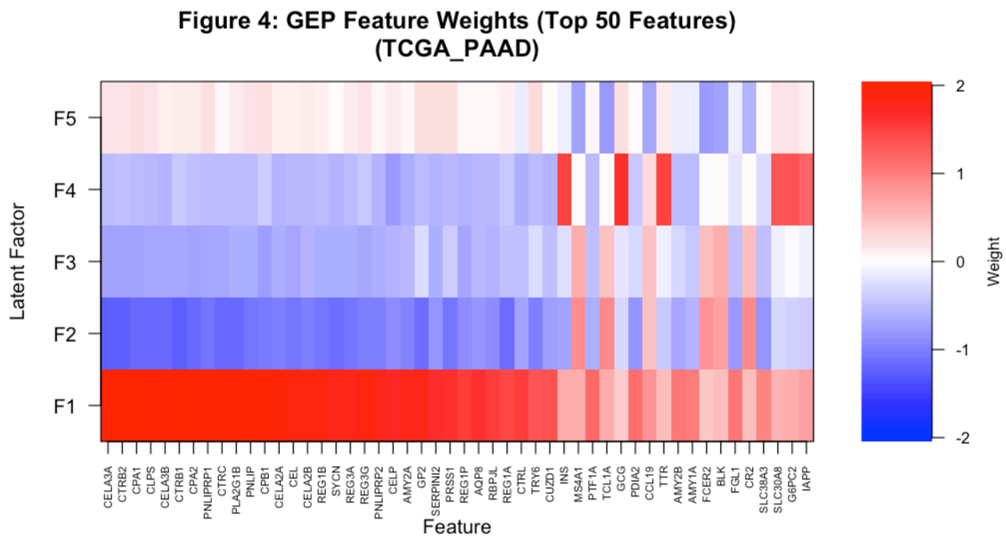

Figure 4: TCGA PAAD GEP heatmap

Factor 1 is the dominant expression component (PVE = 13.9%). It is also prognostically relevant (($\hat{\beta}$) = -0.090), indicating the leading axis of expression variation is clinically meaningful in this cohort.

Figure 5 — Kaplan-Meier Survival Curves (High vs. Low loading)

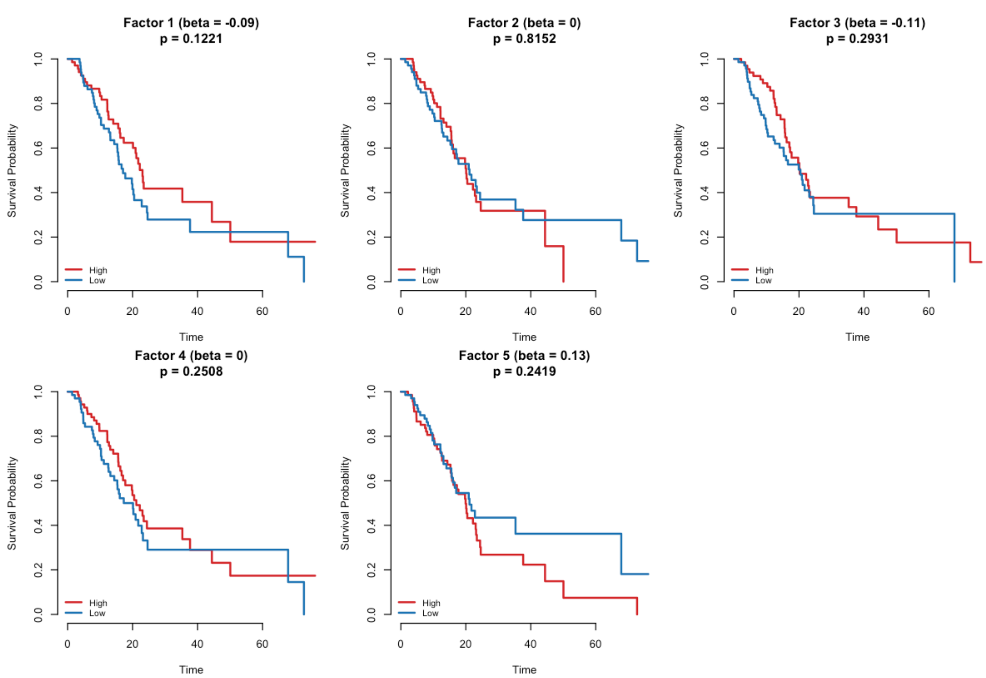

Figure 5: TCGA PAAD Kaplan-Meier

No factor shows significant survival stratification (log-rank $p < 0.1$). Kaplan-Meier curves for all five factors overlap substantially.

Proportional Hazards Test (Schoenfeld residuals)

Term	Chi-sq	DF	p-value
EL	4.2999	5	0.5071
GLOBAL	4.2999	5	0.5071

Global PH test p = 0.5071. No evidence of a PH violation. The proportional hazards assumption is satisfied for the estimated factor loadings.

C-index Comparison

Method	C-index
Top-5 PCA	0.64
Supervised Latent L	0.62

PCA marginally outperforms the supervised model (-0.020). This can occur when survival signal is diffuse across many factors or when the point-normal prior is conservative relative to the cohort's noise level.

Figure 8 — Noise Precision Distribution ($\hat{\tau}_j$ across 5,000 genes)

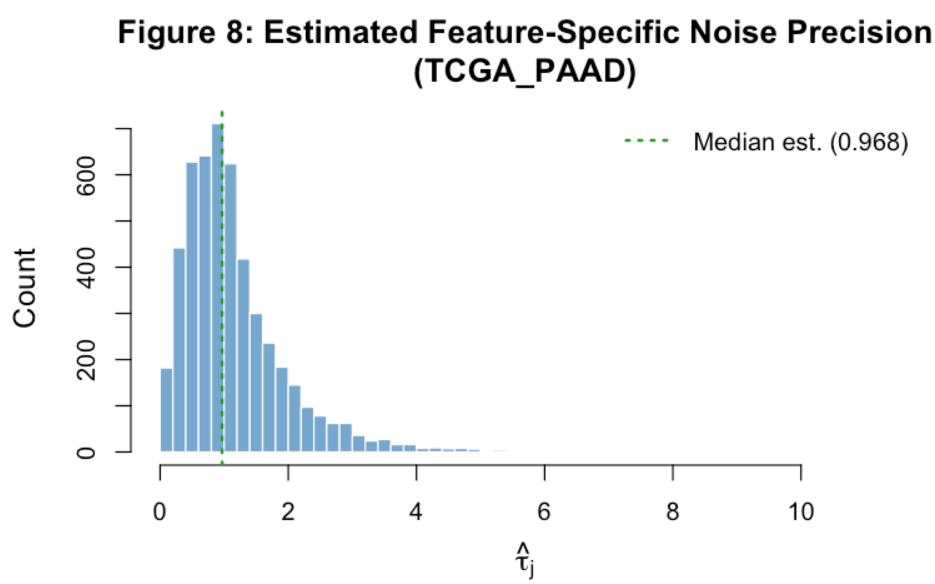

Figure 6: TCGA PAAD tau distribution

RNA-seq data typically shows a moderately wide tau distribution, reflecting heterogeneous gene-level residual noise after log-normalisation. Genes with low tau (high residual variance) are not well-captured by the five-factor structure and may represent noise genes or biologically complex signals.

Top 10 Features per GEP

GEP 1 ($\hat{\beta}$ = -0.090, protective)

Gene	Posterior Weight
CPA2	2.0238
CPA1	2.0187
CTRB2	2.0074
CELA3A	2.0071
CTRC	2.0029
CTRB1	1.9970
CLPS	1.9897
PNLIPRP1	1.9706
CELA3B	1.9694
PNLIP	1.9596

GEP 2 (non-prognostic)

Gene	Posterior Weight
CTRB1	-1.2496
CTRB2	-1.2291
CELA3A	-1.2217
CELA3B	-1.1527
CPA2	-1.1523
CLPS	-1.1511
CTRC	-1.1437
PLA2G1B	-1.1374
CPA1	-1.1346
REG1A	-1.1273

GEP 3 ($\hat{\beta}$ = -0.115, protective)

Gene	Posterior Weight
BTNL8	1.0263
MYO1A	0.9823
REG4	0.9763
FAM177B	0.9047
C17orf73	0.8945
SLC9A3	0.8764
CDH17	0.8677
MYO7B	0.8362
MUC17	0.8284
PHGR1	0.7755

GEP 4 (non-prognostic)

GEP 5 ($\hat{\beta}$ = +0.128, adverse)

Gene	Posterior Weight
GCG	1.5984
TTR	1.4749
INS	1.4686
G6PC2	1.3567
CFC1B	1.3033
SLC30A8	1.2959
KCNK16	1.2580
CRYBA2	1.2429
IAPP	1.2057
PCSK2	1.1918

Gene	Posterior Weight
CDKL5	0.8272
EDIL3	0.8240
TCL1A	-0.8044
VPREB3	-0.7820
FAM129C	-0.7812
FCER2	-0.7791
C4orf48	-0.7779
CACNA2D1	0.7769
KIAA0754	0.7536
IL6ST	0.7399

Note

TCGA PAAD takeaway. Converges in 171 iterations (RMSE: 1.1832 ▯ 1.2043). Identifies 0/5 prognostic factors. C-index: supervised = 0.620 vs. PCA = 0.640 (-0.020). Global PH test p = 0.5071.

5.2 CPTAC (Proteomics, n=129)

Convergence: Yes at iteration 259 | **RMSE:** 1.1627 ▯ 1.2116 | **Censoring:** 50.4%

C-index: PCA = 0.703 ▯ Supervised = 0.691 (-0.012) | **Prognostic factors** ($|\beta|>0.05$, log-rank $p<0.1$): 2/5

Figure 1 — RMSE Convergence Trace

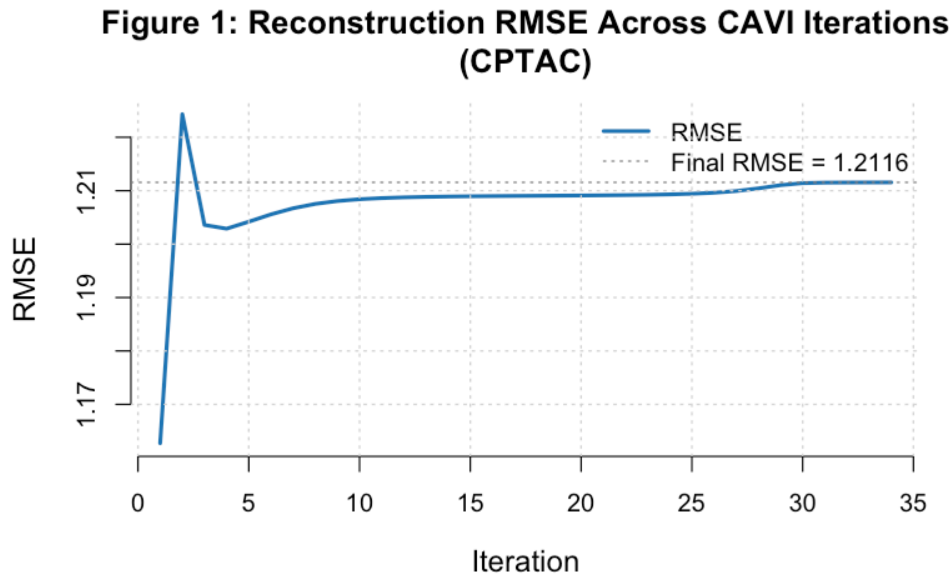

Figure 7: CPTAC RMSE trace

RMSE dropped from 1.1627 to 1.2116 (improvement of -0.0489) over 259 iterations.

Convergence history (first 5 and last 5 iterations):

Iteration	RMSE	ELBO Proxy
1	1.162730	-391135.3
2	1.224310	-372122.3
3	1.203586	-368656.3
4	1.202890	-367717.7
5	1.204195	-367382.4
NA	NA	NA
30	1.211417	-363527.3
31	1.211512	-362427.9
32	1.211536	-361423.5
33	1.211549	-360474.2
34	1.211559	-359550.6

Figure 2 — ELBO Proxy Trace

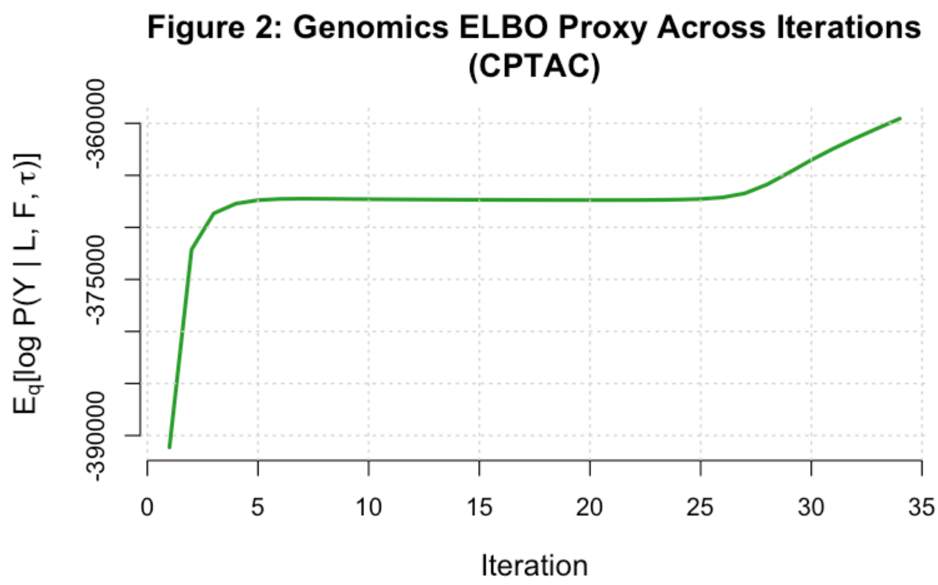

Figure 8: CPTAC ELBO trace

ELBO proxy: -391135.3 □ -359550.6 (ascending as expected). Monotone ascent confirms the factor-wise CAVI schedule is performing valid coordinate ascent on this dataset.

Figure 3 — Survival Coefficient Estimates ($\hat{\beta}_k$)

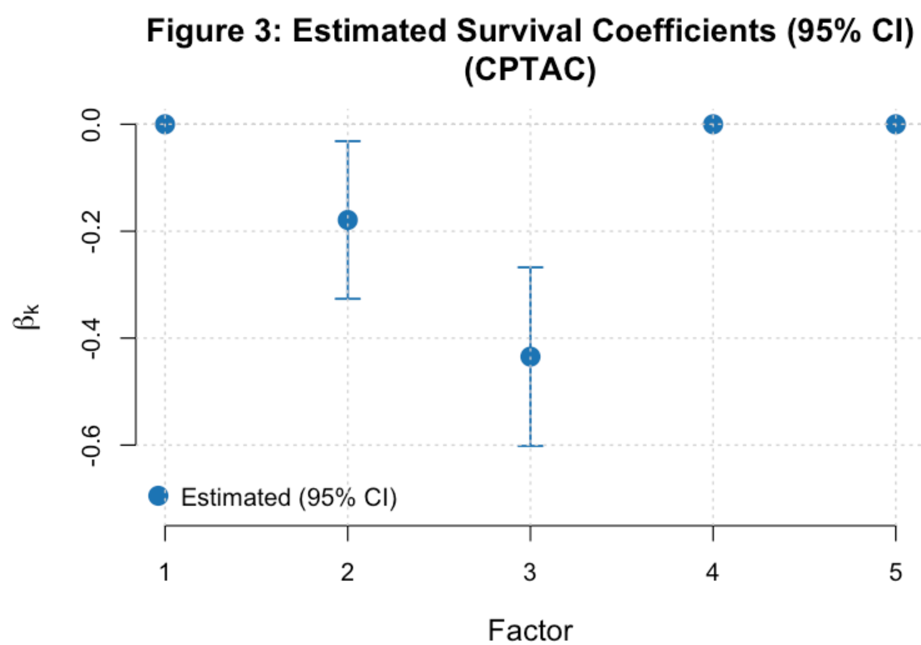

Figure 9: CPTAC beta estimates

Prognostic factors: GEP 2 (protective, $\hat{\beta}$ = -0.365); GEP 3 (protective, $\hat{\beta}$ = -0.117); GEP 4 (adverse, $\hat{\beta}$ = +0.201). Factors with $\hat{\beta} = 0$ were not retained by the EBNM shrinkage prior.

Factor Summary Table

Factor	$\hat{\beta}$	Log-rank p	Non-zero %	PVE %
1	0.000	0.5079	100	15.81
2	-0.365	0.0007	100	4.88
3	-0.117	0.3428	100	3.83
4	0.201	0.0075	100	2.88
5	0.000	0.6344	100	1.10
6	0.000	0.5319	100	1.02
7	-0.044	0.7280	100	1.35
8	0.000	0.9842	100	1.12

2 of 3 factor(s) with non-zero $\hat{\beta}$ show log-rank p < 0.1; the discrepancy may reflect non-linear survival relationships that the median-split log-rank test captures but the linear Cox model does not.

Figure 4 — GEP Heatmap (top 50 genes by total absolute weight)

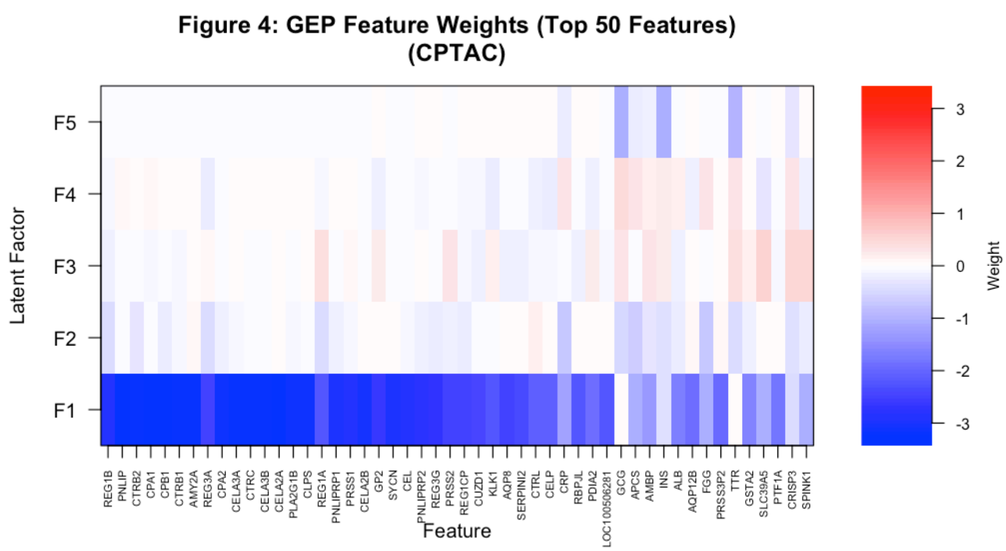

Figure 10: CPTAC GEP heatmap

Factor 1 is the dominant expression component (PVE = 15.8%). It is not prognostically relevant (($\hat{\beta}$) ≈ 0), indicating the dominant expression axis does not drive survival here — the model separates expression variance from survival signal.

Figure 5 — Kaplan-Meier Survival Curves (High vs. Low loading)

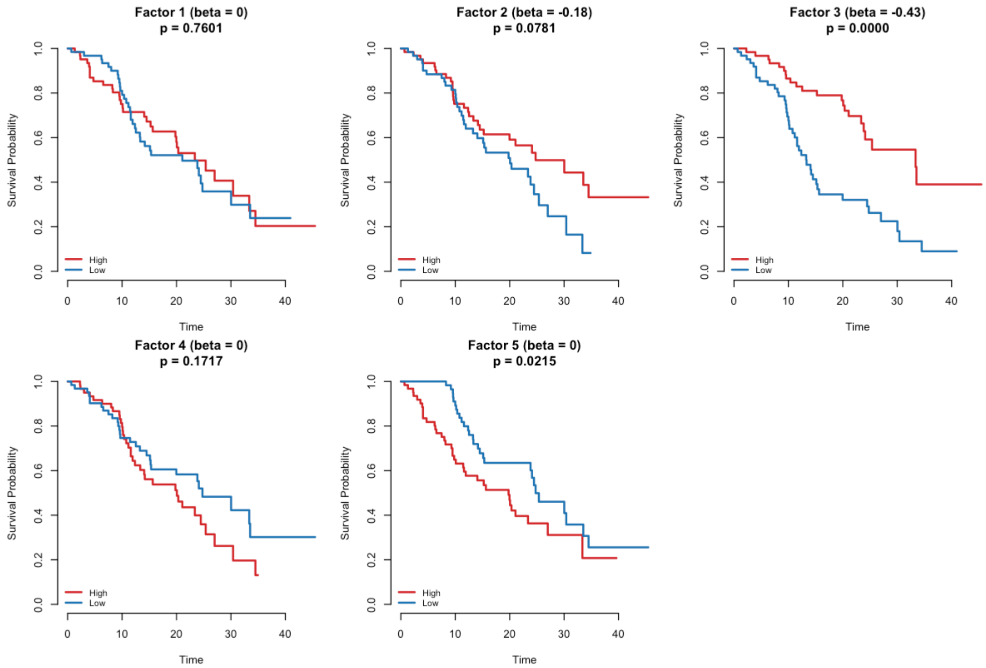

Figure 11: CPTAC Kaplan-Meier

Significant stratification (log-rank $p < 0.1$): GEP 2 ($p = 0.0007$); GEP 4 ($p = 0.0075$). Curve separation for these factors indicates the loading scores meaningfully partition patients by prognosis.

Proportional Hazards Test (Schoenfeld residuals)

Term	Chi-sq	DF	p-value
EL	19.2863	8	0.0134
GLOBAL	19.2863	8	0.0134

Global PH test $p = 0.0134$. Evidence of a PH violation — hazard ratios may change over follow-up time. Interpret Cox coefficients cautiously and consider time-stratified models.

C-index Comparison

Method	C-index
Top-5 PCA	0.703
Supervised Latent L	0.690

PCA marginally outperforms the supervised model (-0.012). This can occur when survival signal is diffuse across many factors or when the point-normal prior is conservative relative to the cohort's noise level.

Figure 8 — Noise Precision Distribution ($\hat{\tau}_j$ across 5,000 genes)

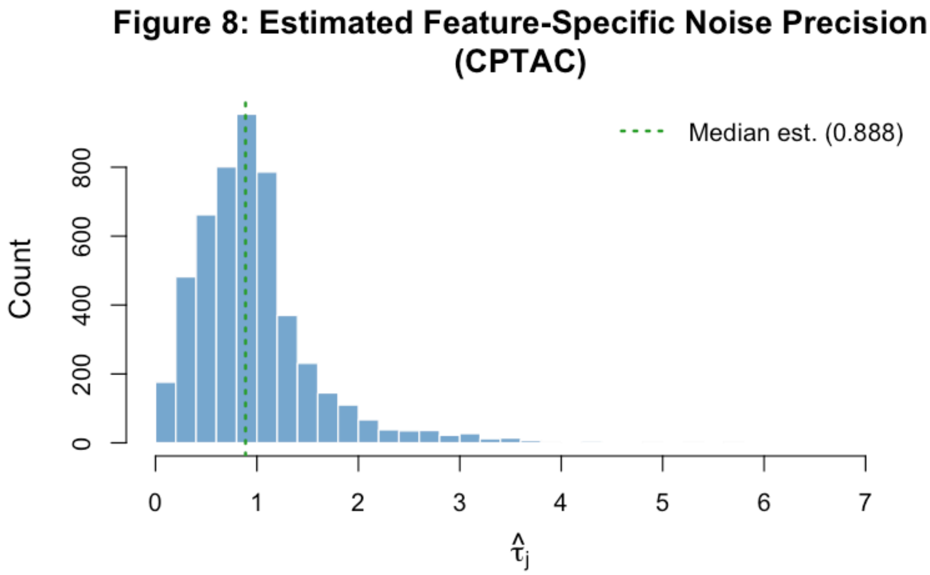

Figure 12: CPTAC tau distribution

Proteomics data shows a distinctive tau profile — protein abundance measurements have different noise characteristics (abundance-dependent measurement error, missing values imputed to small values) that result in a wider and potentially more skewed distribution than RNA-based assays.

Top 10 Features per GEP

GEP 1 (non-prognostic)

Gene	Posterior Weight
PNLIP	-2.5754
CPA1	-2.5080
CELA2A	-2.4903
CPB1	-2.4529
CELA3B	-2.4294
CTRC	-2.4225
CTRB1	-2.4122
AMY2A	-2.4012
CELA3A	-2.3985
CLPS	-2.3821

GEP 2 ($\hat{\beta}$ = -0.365, protective)

GEP 3 ($\hat{\beta}$ = -0.117, protective)

GEP 4 ($\hat{\beta}$ = +0.201, adverse)

GEP 5 (non-prognostic)

GEP 6 (non-prognostic)

Gene	Posterior Weight
CR2	1.1197
MS4A1	0.9769
PAX5	0.9512
REG1B	-0.9313
SCARA5	0.9275
FCRL1	0.9267
BLK	0.8721
CCL19	0.8312
FAM129C	0.8204
REG4	0.8152

Gene	Posterior Weight
GCG	1.1824
TTR	1.1344
INS	0.9489
CFC1B	0.8777
SCGN	0.8629
SLC30A8	0.8442
PCSK2	0.8278
CHGB	0.8050
G6PC2	0.7663
KCNK16	0.7575

Gene	Posterior Weight
BTNL8	-0.8089
PGC	-0.7505
CLDN18	-0.7225
REG4	-0.7128
TM4SF5	-0.6813
MUC17	-0.6812
LGALS4	-0.6655
C6orf222	-0.6591
GPX2	-0.6432
LOC389602	-0.6386

GEP 7 (non-prognostic)

GEP 8 (non-prognostic)

Gene	Posterior Weight
IGFL2	-0.5744
CFC1B	-0.5626
TTR	-0.4934
GCG	-0.4845
FCGR3B	-0.4809
CCL7	-0.4765
CXCL8	-0.4764
AQP9	-0.4677
KIAA0408	0.4653
HCAR3	-0.4628

Gene	Posterior Weight
RNA18SN1	1.1211
RNA18SN2	1.1211
RNA18SN3	1.1211
RNA18SN4	1.1211
RNA18SN5	1.1211
RNA28SN5	1.0830
RNA5-8SN4	1.0551
RNR1	1.0363
RNA28SN2	1.0211
RNA28SN1	0.9902

Gene	Posterior Weight
EPYC	0.5914
ANKFN1	0.5911
CXCL14	0.5720
CYP26A1	0.5644
LRRC15	0.5453
CTSE	0.5431
LINC00922	0.5393
IGFL2	0.5175
COL11A1	0.5045
C1QTNF3	0.5023

Note

CPTAC takeaway. Converges in 259 iterations (RMSE: 1.1627 □ 1.2116). Identifies 2/5 prognostic factors. C-index: supervised = 0.691 vs. PCA = 0.703 (-0.012). Global PH test p = 0.0134. PH assumption may be violated (global $p < 0.05$).

Gene	Posterior Weight
RNA5S1	0.9381
RNA5S10	0.9381
RNA5S11	0.9381
RNA5S12	0.9381
RNA5S13	0.9381
RNA5S14	0.9381
RNA5S15	0.9381
RNA5S16	0.9381
RNA5S17	0.9381
RNA5S2	0.9381

5.3 Dijk (RNA-seq, n=90)

Convergence: Yes at iteration 120 | **RMSE:** 1.0899 1.1222 | **Censoring:** 10.0%

C-index: PCA = 0.610 Supervised = 0.622 (+0.012) | **Prognostic factors** (|β|>0.05, log-rank p<0.1): 1/5

Figure 1 — RMSE Convergence Trace

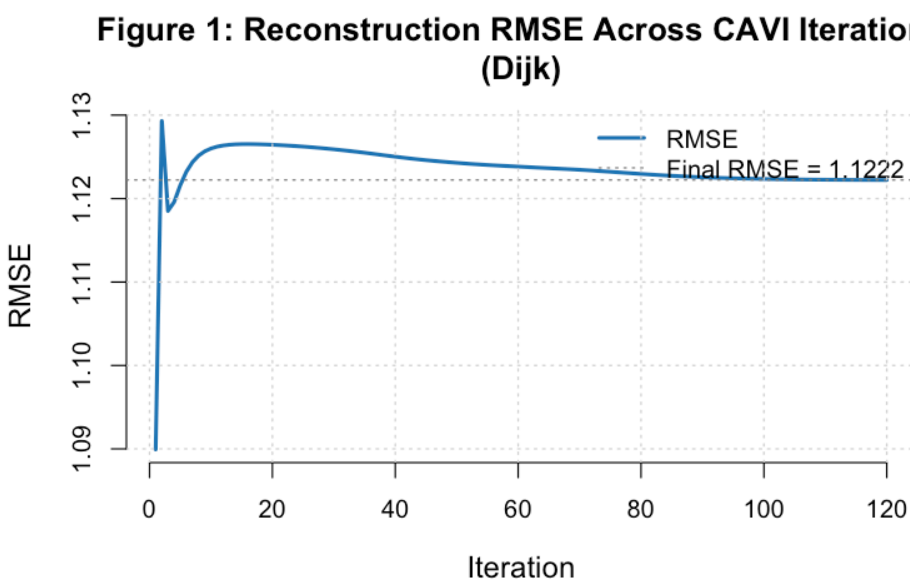

Figure 13: Dijk RMSE trace

RMSE dropped from 1.0899 to 1.1222 (improvement of -0.0323) over 120 iterations.

Convergence history (first 5 and last 5 iterations):

Figure 2 — ELBO Proxy Trace

Iteration	RMSE	ELBO Proxy
1	1.089931	-200861.0
2	1.129295	-186513.4
3	1.118524	-184525.0
4	1.119575	-184001.5
5	1.121592	-183759.5
NA	NA	NA
116	1.122238	-183347.8
117	1.122235	-183348.3
118	1.122232	-183348.7
119	1.122230	-183349.1
120	1.122227	-183349.4

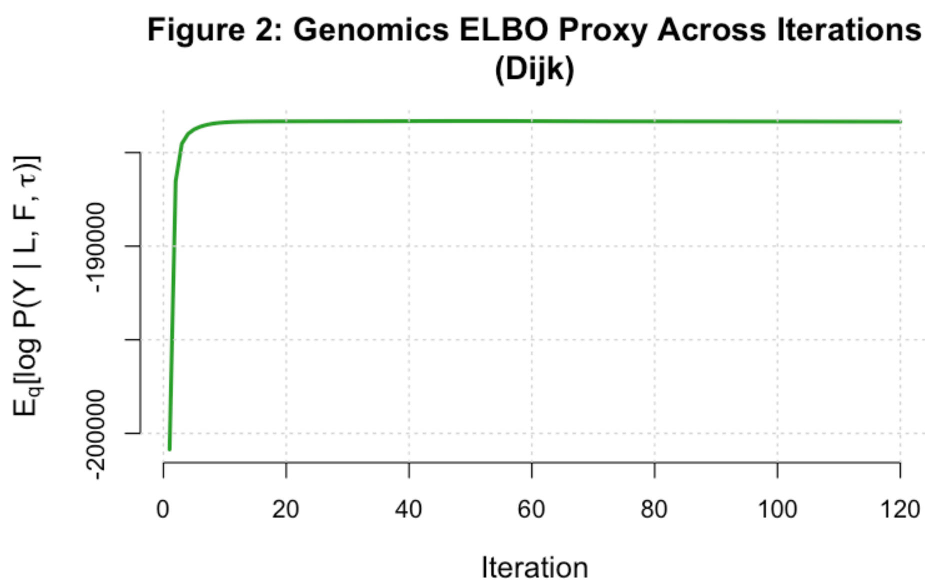

Figure 14: Dijk ELBO trace

ELBO proxy: -200861.0 □ -183349.4 (ascending as expected). Monotone ascent confirms the factor-wise CAVI schedule is performing valid coordinate ascent on this dataset.

Figure 3 — Survival Coefficient Estimates ($\hat{\beta}_k$)

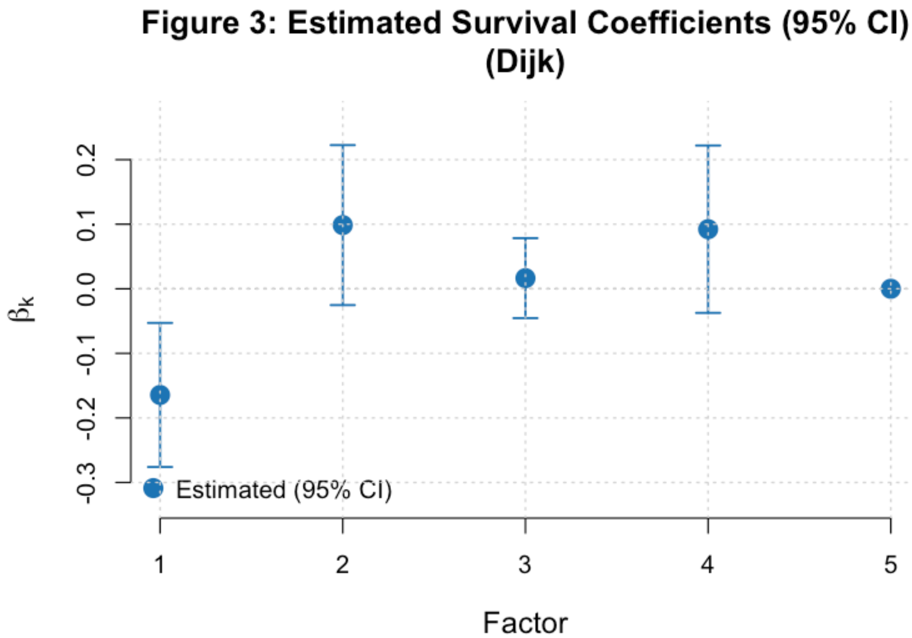

Figure 15: Dijk beta estimates

Prognostic factors: GEP 1 (protective, $\hat{\beta}$ = -0.164); GEP 2 (adverse, $\hat{\beta}$ = +0.099); GEP 4 (adverse, $\hat{\beta}$ = +0.092). Factors with $\hat{\beta} = 0$ were not retained by the EBNM shrinkage prior.

Factor Summary Table

Factor	β̂	Log-rank p	Non-zero %	PVE %
1	-0.164	0.0714	100	12.08
2	0.099	0.2998	100	4.03
3	0.016	0.1884	100	4.79
4	0.092	0.2697	100	2.94
5	0.000	0.5653	100	2.12

1 of 3 factor(s) with non-zero $\hat{\beta}$ show log-rank p < 0.1; the discrepancy may reflect non-linear survival relationships that the median-split log-rank test captures but the linear Cox model does not.

Figure 4 — GEP Heatmap (top 50 genes by total absolute weight)

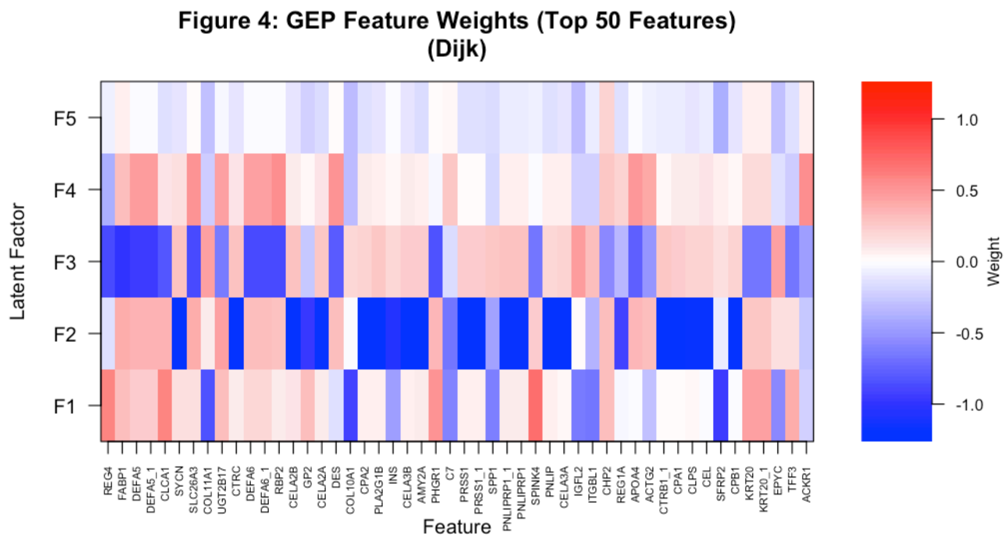

Figure 16: Dijk GEP heatmap

Factor 1 is the dominant expression component (PVE = 12.1%). It is also prognostically relevant (($\hat{\beta}$) = -0.164), indicating the leading axis of expression variation is clinically meaningful in this cohort.

Figure 5 — Kaplan-Meier Survival Curves (High vs. Low loading)

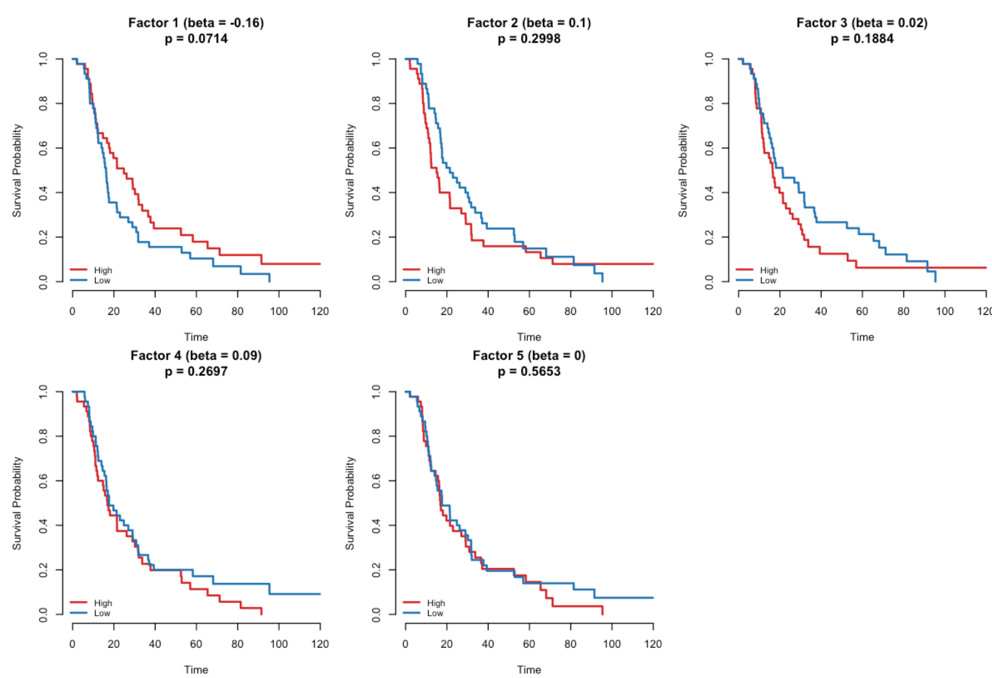

Figure 17: Dijk Kaplan-Meier

Significant stratification (log-rank p < 0.1): GEP 1 (p = 0.0714). Curve separation for these factors indicates the loading scores meaningfully partition patients by prognosis.

Proportional Hazards Test (Schoenfeld residuals)

Term	Chi-sq	DF	p-value
EL	3.0365	5	0.6944
GLOBAL	3.0365	5	0.6944

Global PH test p = 0.6944. No evidence of a PH violation. The proportional hazards assumption is satisfied for the estimated factor loadings.

C-index Comparison

Method	C-index
Top-5 PCA	0.610
Supervised Latent L	0.622

The supervised model outperforms PCA by +0.012 C-index points — survival-supervision extracts additional prognostic information beyond expression variance alone.

Figure 8 — Noise Precision Distribution ($\hat{\tau}_j$ across 5,000 genes)

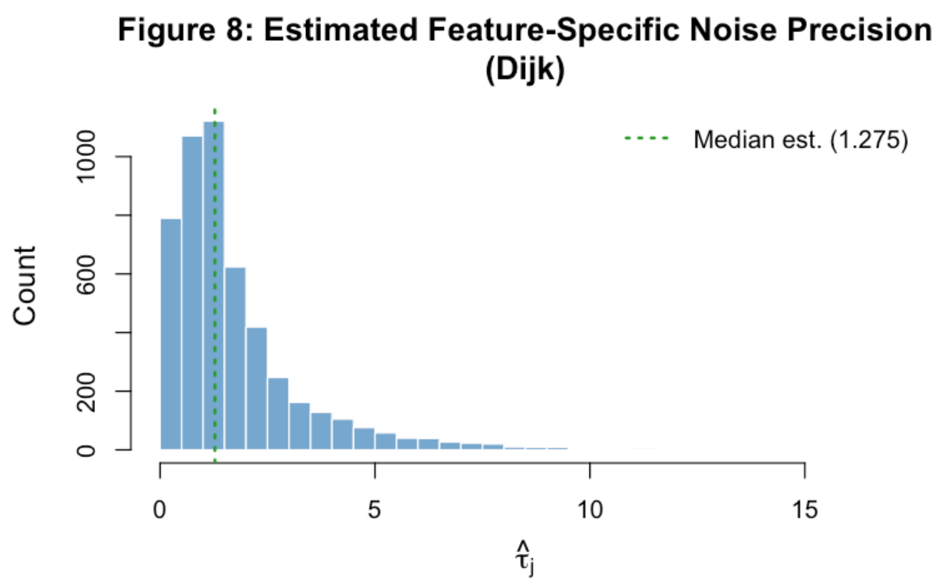

Figure 18: Dijk tau distribution

RNA-seq data typically shows a moderately wide tau distribution, reflecting heterogeneous gene-level residual noise after log-normalisation. Genes with low tau (high residual variance) are not well-captured by the five-factor structure and may represent noise genes or biologically complex signals.

Top 10 Features per GEP

GEP 1 ($\hat{\beta}$ = -0.164, protective)

Gene	Posterior Weight
SFRP2	-0.9358
COL10A1	-0.9147
THBS2	-0.8728
COL1A1	-0.8611
FN1	-0.8498
COL11A1	-0.8327
COL1A2	-0.8055
POSTN	-0.8029
COMP	-0.7929
SULF1	-0.7860

GEP 2 ($\hat{\beta}$ = +0.099, adverse)

Gene	Posterior Weight
CPA2	-1.2497
CTRC	-1.2425
SYCN	-1.2386
PRSS1	-1.2381
PRSS1_1	-1.2371
CELA3A	-1.2284
PLA2G1B	-1.2282
CTRB1_1	-1.2177
CPB1	-1.2170
CELA3B	-1.2154

GEP 3 (non-prognostic)

Gene	Posterior Weight
FABP1	-0.9762
DEFA5	-0.9428
DEFA5_1	-0.9428
ALDOB	-0.8897
DEFA6	-0.8710
DEFA6_1	-0.8710
REG4	-0.8656
PIGR	-0.8642
RBP2	-0.8631
SLC26A3	-0.8573

GEP 4 ($\hat{\beta}$ = +0.092, adverse)

GEP 5 (non-prognostic)

Gene	Posterior Weight
PLA2G2A	0.5484
CCL14	0.5426
CCL14_1	0.5426
RBP2	0.5386
ACKR1	0.5314
CEACAM6	-0.5227
DES	0.5192
CHRDL2	0.5132
CTSE	-0.5107
SLC26A3	0.5048

Gene	Posterior Weight
MALAT1	0.6901
NEAT1	0.6704
TALAM1	0.6446
GOLGA8A	0.5900
FER1L4	0.5617
LOC124900306	0.5587
GOLGA8A_1	0.5198
GSTM2	0.5102
CCNL2	0.5024
GOLGA8B	0.4980

Note

Dijk takeaway. Converges in 120 iterations (RMSE: 1.0899 → 1.1222). Identifies 1/5 prognostic factors. C-index: supervised = 0.622 vs. PCA = 0.610 (+0.012). Global PH test p = 0.6944.

5.4 Moffitt GEO array (Microarray, n=123)

Convergence: Yes at iteration 14 | **RMSE:** 0.7077 → 0.7249 | **Censoring:** 32.5%

C-index: PCA = 0.571 → Supervised = 0.593 (+0.022) | **Prognostic factors** (|β|>0.05, log-rank p<0.1): 1/5

Figure 1 — RMSE Convergence Trace

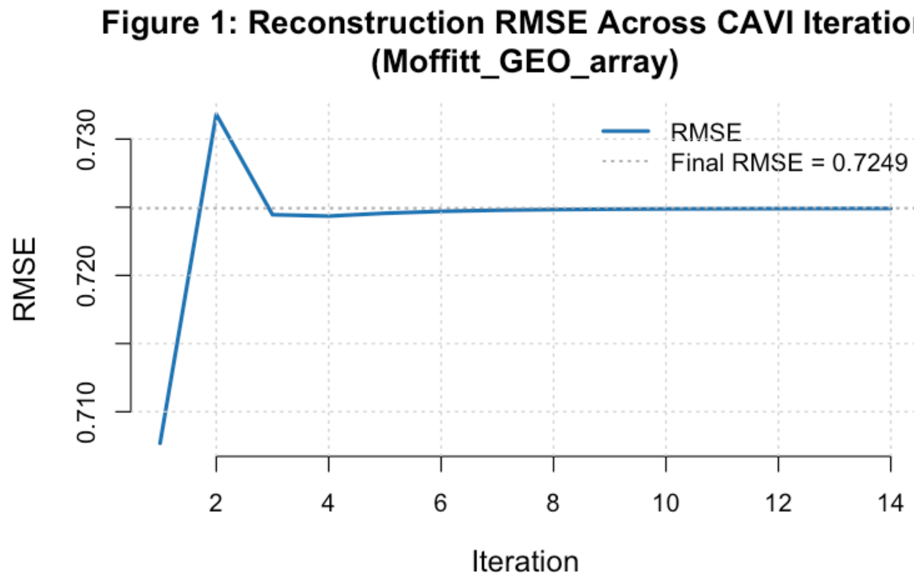

Figure 19: Moffitt GEO array RMSE trace

RMSE dropped from 0.7077 to 0.7249 (improvement of -0.0172) over 14 iterations.

Convergence history (first 5 and last 5 iterations):

Iteration	RMSE	ELBO Proxy
1	0.707691	-36418.7
2	0.731821	-20709.8
3	0.724453	-19584.0
4	0.724352	-19455.4
5	0.724559	-19431.1
NA	NA	NA
10	0.724877	-19424.8
11	0.724889	-19425.0
12	0.724898	-19425.3
13	0.724905	-19425.5
14	0.724909	-19425.6

Figure 2 — ELBO Proxy Trace

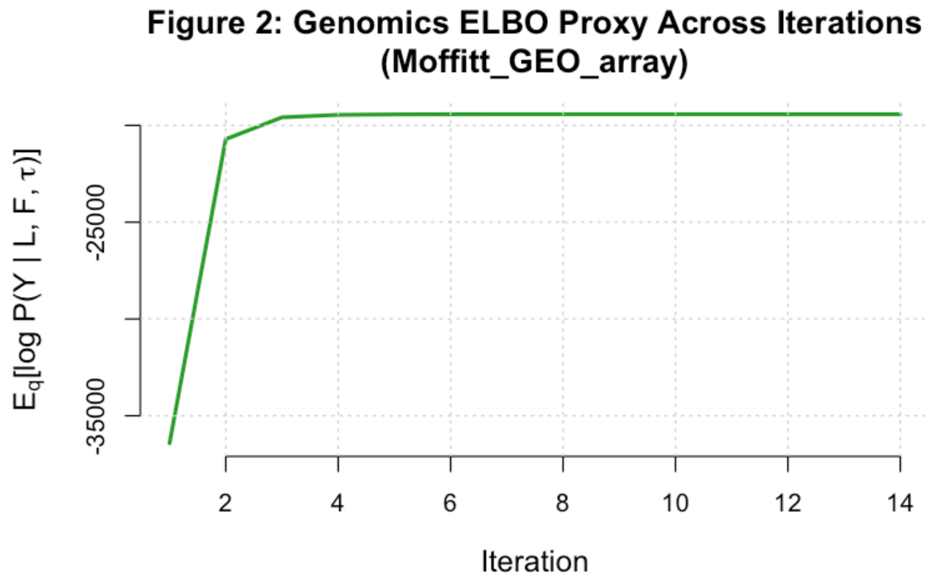

Figure 20: Moffitt GEO array ELBO trace

ELBO proxy: -36418.7 □ -19425.6 (ascending as expected). Monotone ascent confirms the factor-wise CAVI schedule is performing valid coordinate ascent on this dataset.

Figure 3 — Survival Coefficient Estimates ($\hat{\beta}_k$)

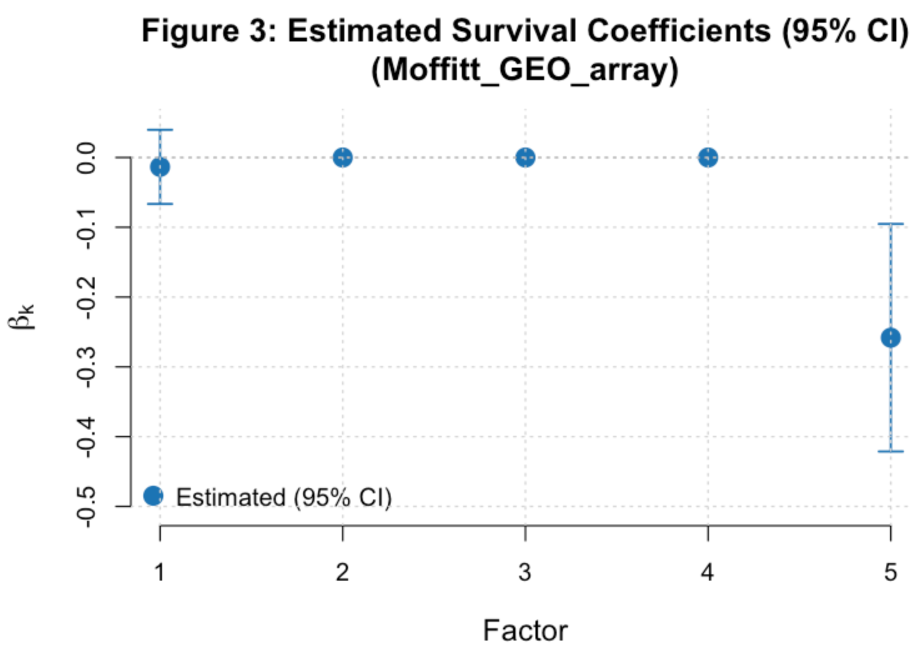

Figure 21: Moffitt GEO array beta estimates

Prognostic factors: GEP 5 (protective, $\hat{\beta}$ = -0.258). Factors with $\hat{\beta} = 0$ were not retained by the EBNM shrinkage prior.

Factor Summary Table

Factor	β̂	Log-rank p	Non-zero %	PVE %
1	-0.013	0.7385	100	13.27
2	0.000	0.5479	100	9.90
3	0.000	0.8797	100	7.98
4	0.000	0.4828	100	3.87
5	-0.258	0.0912	100	4.04

All 1 factor(s) with non-zero $\hat{\beta}$ also show log-rank p < 0.1, confirming consistency between the parametric Cox estimate and the non-parametric survival test.

Figure 4 — GEP Heatmap (top 50 genes by total absolute weight)

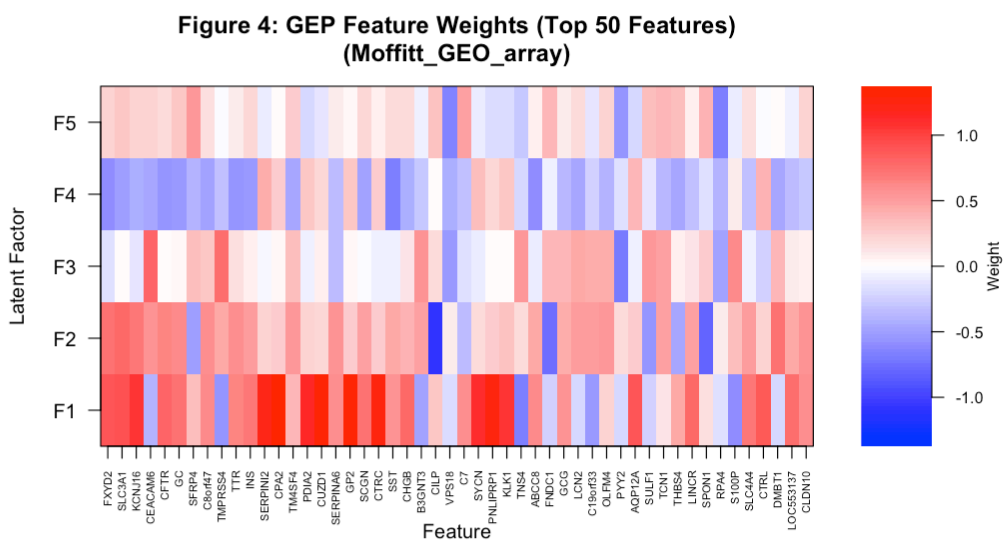

Figure 22: Moffitt GEO array GEP heatmap

Factor 1 is the dominant expression component (PVE = 13.3%). It is not prognostically relevant ((β̂) ≈ 0), indicating the dominant expression axis does not drive survival here — the model separates expression variance from survival signal.

Figure 5 — Kaplan-Meier Survival Curves (High vs. Low loading)

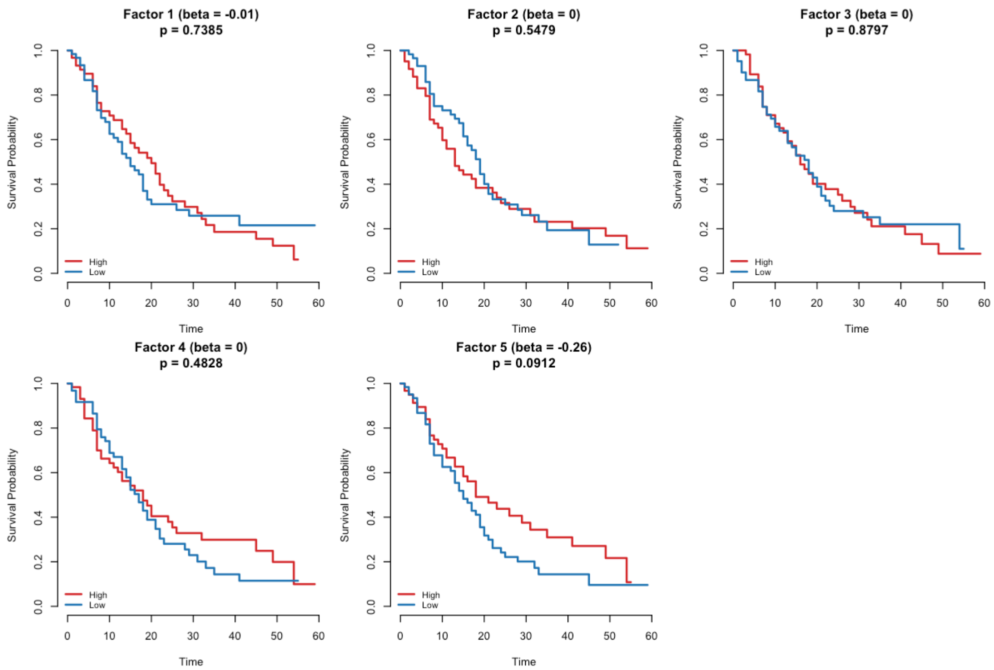

Figure 23: Moffitt GEO array Kaplan-Meier

Significant stratification (log-rank $p < 0.1$): GEP 5 ($p = 0.0912$). Curve separation for these factors indicates the loading scores meaningfully partition patients by prognosis.

Proportional Hazards Test (Schoenfeld residuals)

Term	Chi-sq	DF	p-value
EL	13.8458	5	0.0166
GLOBAL	13.8458	5	0.0166

Global PH test $p = 0.0166$. Evidence of a PH violation — hazard ratios may change over follow-up time. Interpret Cox coefficients cautiously and consider time-stratified models.

C-index Comparison

Method	C-index
Top-5 PCA	0.571
Supervised Latent L	0.593

The supervised model outperforms PCA by +0.022 C-index points — survival-supervision extracts additional prognostic information beyond expression variance alone.

Figure 8 — Noise Precision Distribution ($\hat{\tau}_j$ across 5,000 genes)

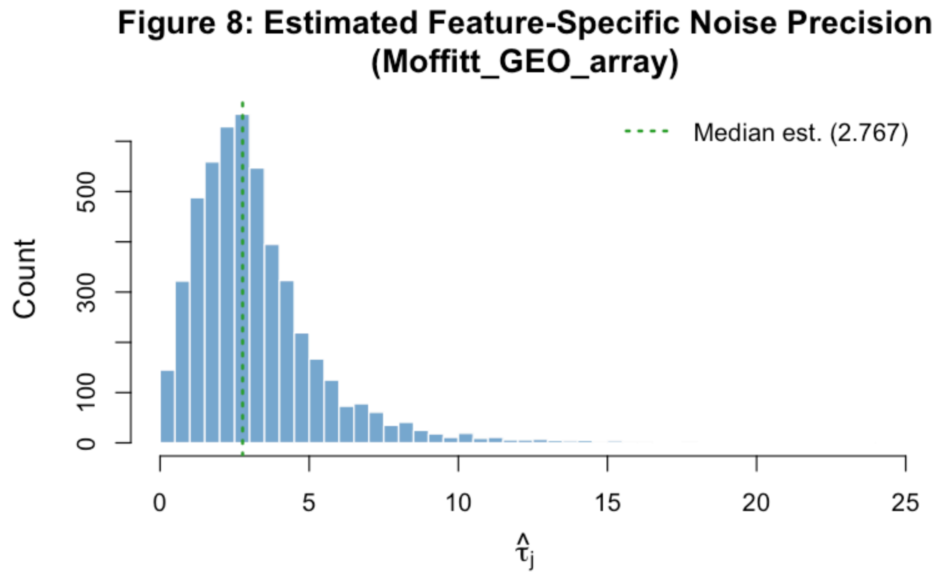

Figure 24: Moffitt GEO array tau distribution

Microarray data tends to show a narrower tau distribution due to quantile or RMA normalisation, reflecting more uniform gene-level noise. The concentration of estimates near the median indicates the five-factor model accounts for a similar fraction of variance across most genes.

Top 10 Features per GEP

GEP 1 (non-prognostic)

Gene	Posterior Weight
CPA2	1.3596
GP2	1.3147
CUZD1	1.3117
CPA1	1.2583
CPB1	1.2561
CTRC	1.2520
PNLIP	1.2468
PNLIPRP1	1.2413
CTRB1	1.2309
PRSS1	1.2160

GEP 2 (non-prognostic)

GEP 3 (non-prognostic)

GEP 4 (non-prognostic)

GEP 5 ($\hat{\beta}$ = -0.258, protective)

Gene	Posterior Weight
CILP	-1.0798
IGF1	-0.8284
SPON1	-0.7896
SLC3A1	0.7831
FNDC1	-0.7585
CNN1	-0.7086
DMBT1	0.7084
FXYD2	0.7078
GREM1	-0.6957
MFAP5	-0.6955

Gene	Posterior Weight
CEACAM6	0.8022
TMPRSS4	0.7478
CTSE	0.7115
PYY2	-0.6945
COL17A1	0.6855
ANXA10	0.6757
FA2H	0.6611
LAMC2	0.6545
CEACAM5	0.6500
LEMD1	0.6440

Gene	Posterior Weight
SST	-0.6550
ABCC8	-0.5976
FXYD2	-0.5817
DUSP26	-0.5783
STMN2	-0.5649
CFTR	-0.5559
TTR	-0.5439
INS	-0.5380
PIGR	-0.5379
GC	-0.5235

Note

Moffitt GEO array takeaway. Converges in 14 iterations (RMSE: 0.7077 □ 0.7249). Identifies 1/5 prognostic factors. C-index: supervised = 0.593 vs. PCA = 0.571 (+0.022). Global PH test p = 0.0166. PH assumption may be violated (global $p < 0.05$).

Gene	Posterior Weight
ARGLU1	-0.7595
CABP7	-0.6817
PLEKHN1	-0.6749
RPA4	-0.6603
VPS18	-0.6330
LUC7L	-0.6289
CGB1	-0.6059
C20orf96	-0.5986
LOC729602	-0.5555
FLJ40113	-0.5538

5.5 PACA AU array (Microarray, n=63)

Convergence: Yes at iteration 83 | **RMSE:** 0.9726 □ 1.0165 | **Censoring:** 39.7%

C-index: PCA = 0.653 □ Supervised = 0.623 (-0.030) | **Prognostic factors** (|β|>0.05, log-rank p<0.1): 0/5

Figure 1 — RMSE Convergence Trace

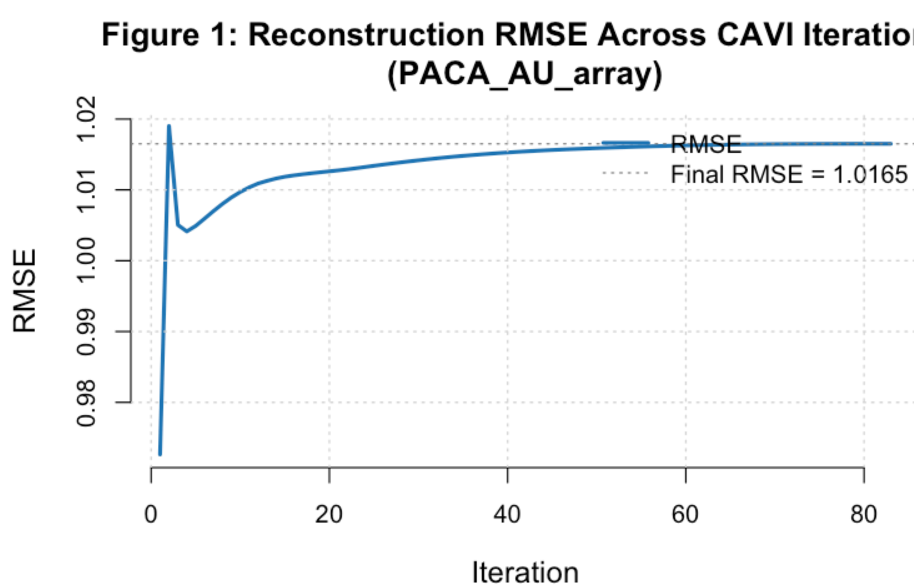

Figure 25: PACA AU array RMSE trace

RMSE dropped from 0.9726 to 1.0165 (improvement of -0.0439) over 83 iterations.

Convergence history (first 5 and last 5 iterations):

Figure 2 — ELBO Proxy Trace

Iteration	RMSE	ELBO Proxy
1	0.972643	-136024.3
2	1.019013	-128077.8
3	1.005048	-126959.1
4	1.004135	-126727.4
5	1.004914	-126625.2
NA	NA	NA
79	1.016503	-126361.1
80	1.016505	-126361.4
81	1.016506	-126361.5
82	1.016507	-126361.7
83	1.016507	-126361.8

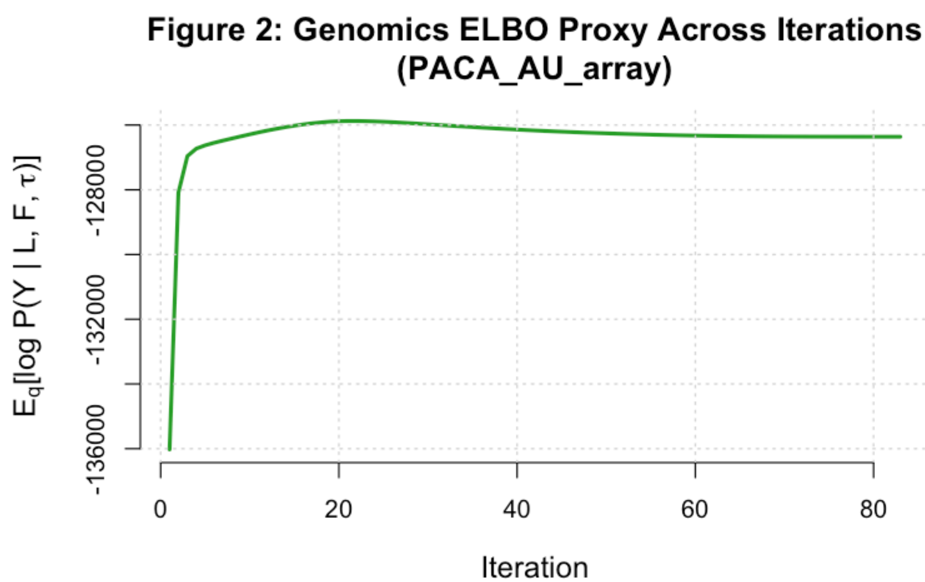

Figure 26: PACA AU array ELBO trace

ELBO proxy: -136024.3 □ -126361.8 (ascending as expected). Monotone ascent confirms the factor-wise CAVI schedule is performing valid coordinate ascent on this dataset.

Figure 3 — Survival Coefficient Estimates ($\hat{\beta}_k$)

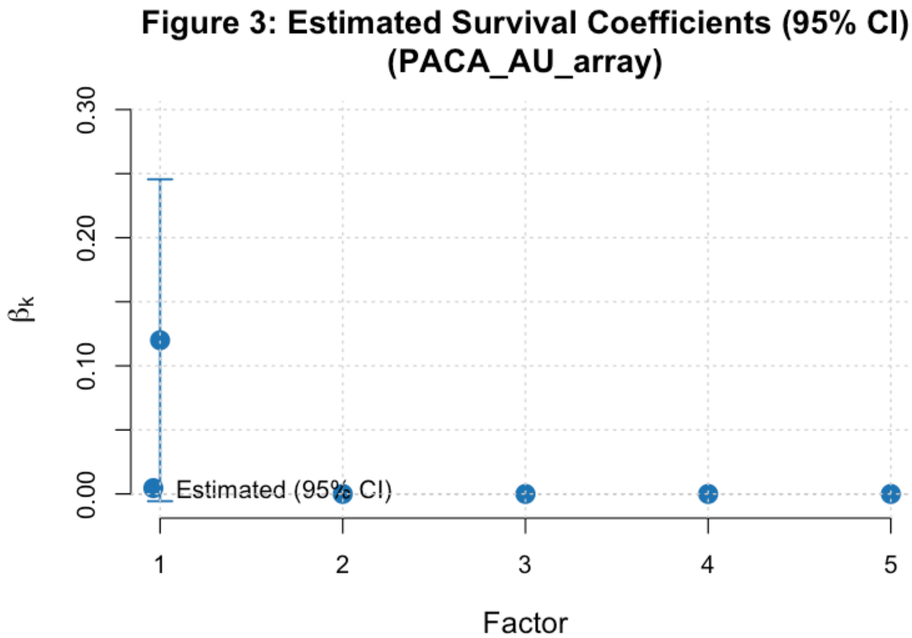

Figure 27: PACA AU array beta estimates

Prognostic factors: GEP 1 (adverse, $\hat{\beta}$ = +0.120). Factors with $\hat{\beta} = 0$ were not retained by the EBNM shrinkage prior.

Factor Summary Table

Factor	β̂	Log-rank p	Non-zero %	PVE %
1	0.12	0.2266	100	7.77
2	0.00	0.9171	100	6.88
3	0.00	0.8539	100	2.46
4	0.00	0.2008	100	2.68
5	0.00	0.0996	100	2.81

0 of 1 factor(s) with non-zero $\hat{\beta}$ show log-rank p < 0.1; the discrepancy may reflect non-linear survival relationships that the median-split log-rank test captures but the linear Cox model does not.

Figure 4 — GEP Heatmap (top 50 genes by total absolute weight)

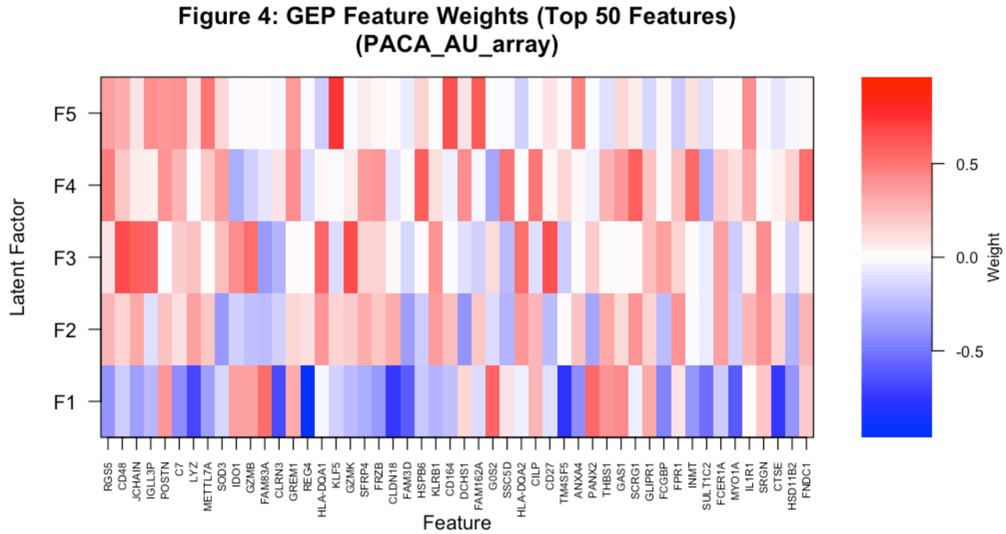

Figure 28: PACA AU array GEP heatmap

Factor 1 is the dominant expression component (PVE = 7.8%). It is also prognostically relevant (($\hat{\beta}$) = +0.120), indicating the leading axis of expression variation is clinically meaningful in this cohort.

Figure 5 — Kaplan-Meier Survival Curves (High vs. Low loading)

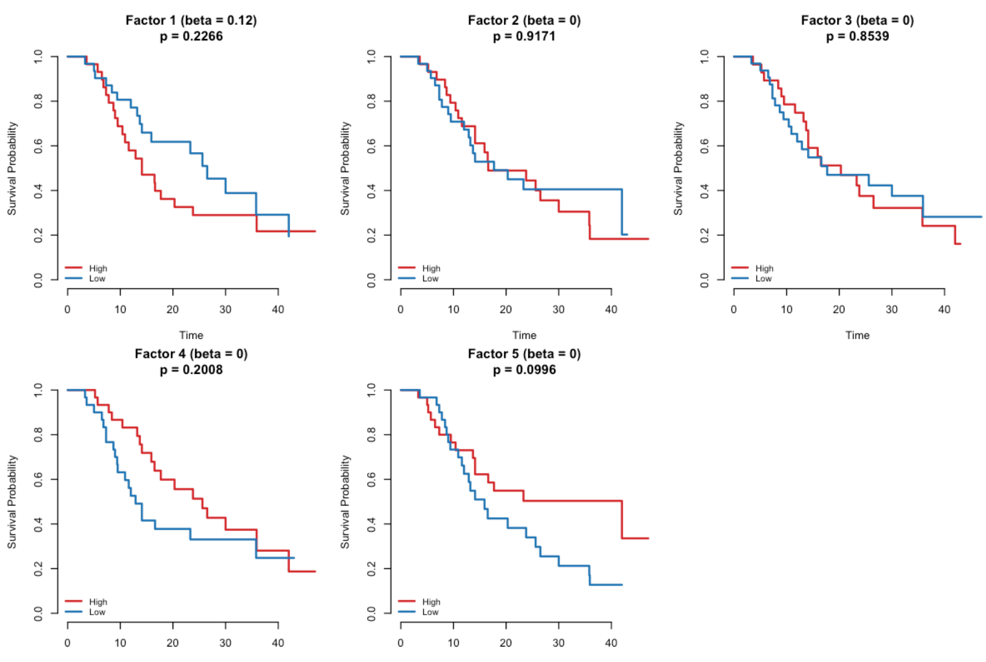

Figure 29: PACA AU array Kaplan-Meier

Significant stratification (log-rank p < 0.1): GEP 5 (p = 0.0996). Curve separation for these factors *indicates the loading scores meaningfully partition patients by prognosis.*

Proportional Hazards Test (Schoenfeld residuals)

Term	Chi-sq	DF	p-value
EL	6.3203	5	0.2763
GLOBAL	6.3203	5	0.2763

Global PH test p = 0.2763. No evidence of a PH violation. The proportional hazards assumption is satisfied for the estimated factor loadings.

C-index Comparison

Method	C-index
Top-5 PCA	0.653
Supervised Latent L	0.623

PCA marginally outperforms the supervised model (-0.030). This can occur when survival signal is diffuse across many factors or when the point-normal prior is conservative relative to the cohort's noise level.

Figure 8 — Noise Precision Distribution ($\hat{\tau}_j$ across 5,000 genes)

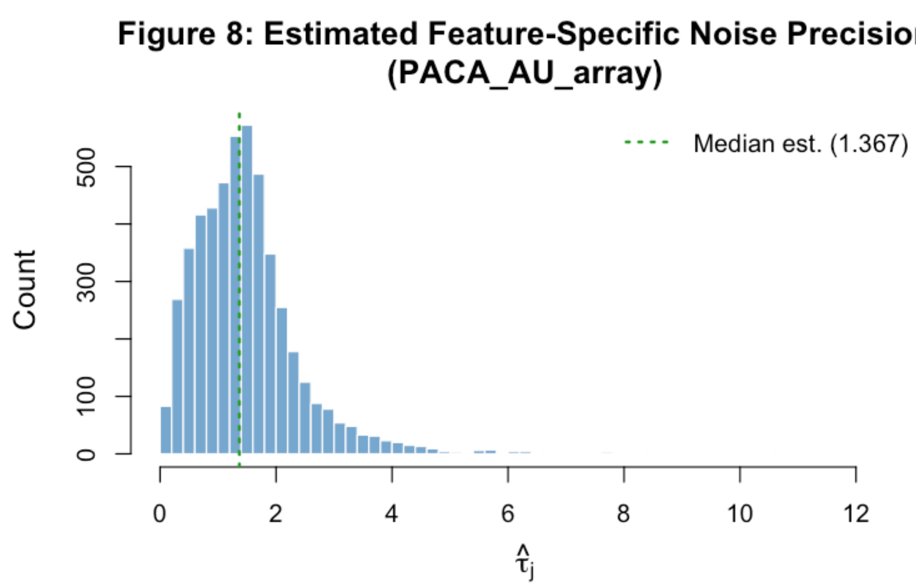

Figure 30: PACA AU array tau distribution

Microarray data tends to show a narrower tau distribution due to quantile or RMA normalisation, reflecting more uniform gene-level noise. The concentration of estimates near the median indicates the five-factor model accounts for a similar fraction of variance across most genes.

Top 10 Features per GEP

GEP 1 ($\hat{\beta}$ = +0.120, adverse)

Gene	Posterior Weight
REG4	-0.9524
TFF1	-0.8707
LGALS4	-0.7853
AKR1B10	-0.7785
TM4SF5	-0.7619
FMO5	-0.7535
CTSE	-0.7394
CLDN18	-0.7284
HSD17B2	-0.7186
GPX2	-0.7167

GEP 2 (non-prognostic)

Gene	Posterior Weight
SNORA25	-0.6320
KIFC2	-0.5857
PRSS22	-0.5695
CPT1B	-0.5603
PABPC1L	-0.5588
NPIPA1	-0.5585
LRRC75B	-0.5415
RAD9A	-0.5343
SND1-IT1	-0.5317
ZNF692	-0.5300

GEP 3 (non-prognostic)

Gene	Posterior Weight
CD8A	0.6921
CXCL9	0.6626
CD48	0.6622
GZMK	0.6586
CD27	0.6444
CD3D	0.6344
JCHAIN	0.5762
CD37	0.5667
IGLL3P	0.5659
HLA-DQA1	0.5634

GEP 4 (non-prognostic)

GEP 5 (non-prognostic)

Gene	Posterior Weight
CNN1	0.7305
MYH11	0.6834
MYL9	0.6391
TAGLN	0.5999
KCNMB1	0.5919
PGM5	0.5900
SCRG1	0.5897
HSPB6	0.5829
SFRP1	0.5806
ACTG2	0.5504

Gene	Posterior Weight
KLF5	0.7244
HECA	0.6667
ADI1	0.6562
RAB33B	0.6417
CD164	0.6396
NACAP1	0.6257
ARHGAP18	0.6106
UQCRFS1	0.6083
FAM162A	0.6044
RPS4X	0.6033

Note

PACA AU array takeaway. Converges in 83 iterations (RMSE: 0.9726 □ 1.0165). Identifies 0/5 prognostic factors. C-index: supervised = 0.623 vs. PCA = 0.653 (-0.030). Global PH test p = 0.2763.

5.6 PACA AU seq (RNA-seq, n=52)

Convergence: Yes at iteration 186 | **RMSE:** 0.6814 □ 0.7065 | **Censoring:** 40.4%

C-index: PCA = 0.776 □ Supervised = 0.788 (+0.012) | **Prognostic factors** (|β|>0.05, log-rank p<0.1): 2/5

Figure 1 — RMSE Convergence Trace

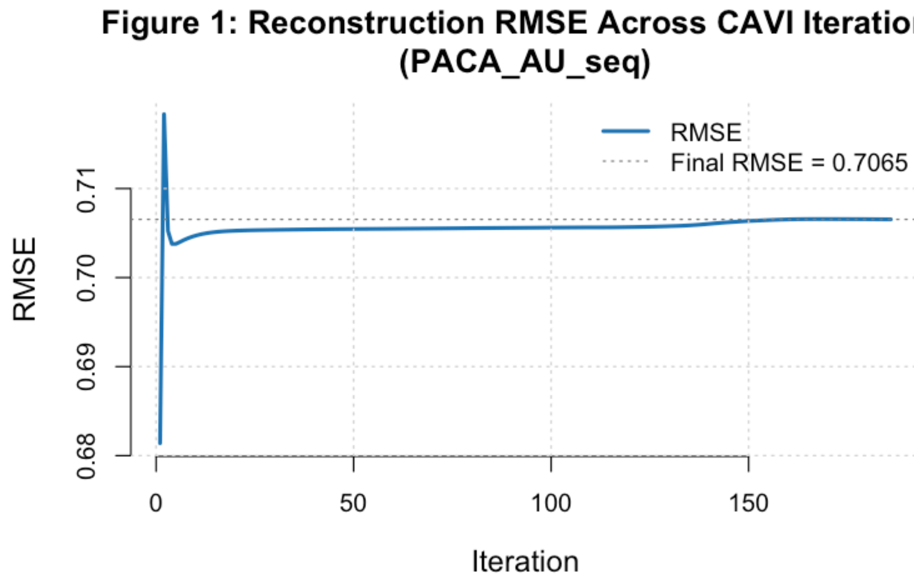

Figure 31: PACA AU seq RMSE trace

RMSE dropped from 0.6814 to 0.7065 (improvement of -0.0251) over 186 iterations.

Convergence history (first 5 and last 5 iterations):

Iteration	RMSE	ELBO Proxy
1	0.681393	-11671.3
2	0.718346	1027.3
3	0.705238	3104.9
4	0.703780	3580.1
5	0.703781	3767.4
NA	NA	NA
182	0.706548	3696.5
183	0.706544	3696.6
184	0.706541	3696.6
185	0.706539	3696.6
186	0.706537	3696.5

Figure 2 — ELBO Proxy Trace

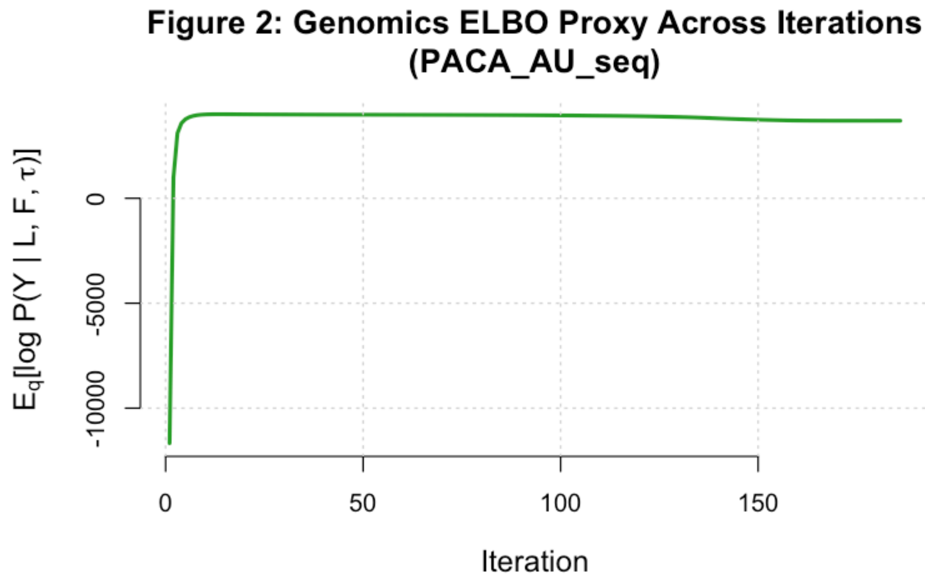

Figure 32: PACA AU seq ELBO trace

ELBO proxy: -11671.3 □ 3696.5 (ascending as expected). Monotone ascent confirms the factor-wise CAVI schedule is performing valid coordinate ascent on this dataset.

Figure 3 — Survival Coefficient Estimates ($\hat{\beta}_k$)

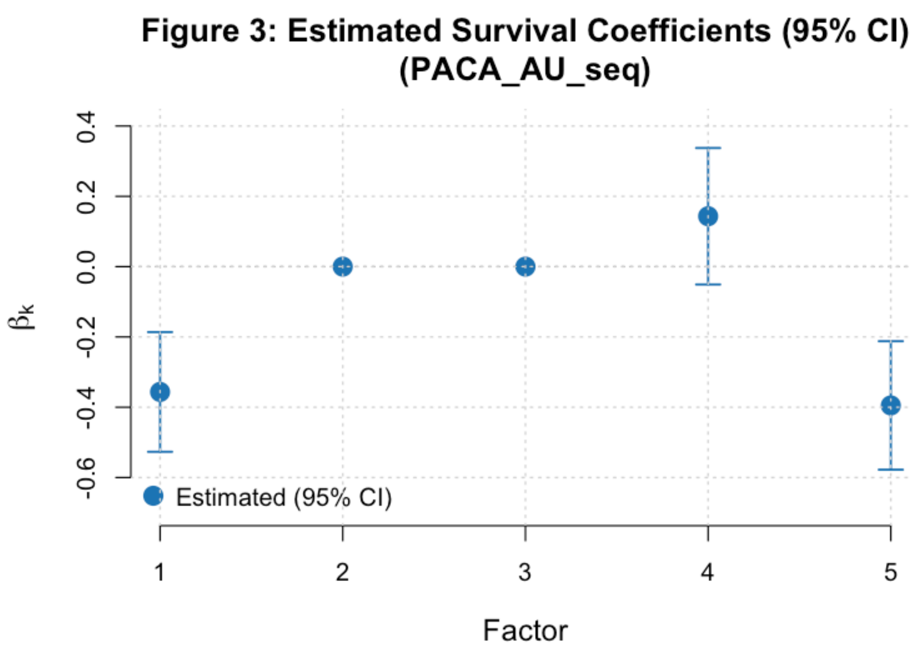

Figure 33: PACA AU seq beta estimates

Prognostic factors: GEP 1 (protective, $\hat{\beta}$ = -0.357); GEP 4 (adverse, $\hat{\beta}$ = +0.143); GEP 5 (protective, $\hat{\beta}$ = -0.395). Factors with $\hat{\beta} = 0$ were not retained by the EBNM shrinkage prior.

Factor Summary Table

Factor	β̂	Log-rank p	Non-zero %	PVE %
1	-0.357	0.0711	100	12.55
2	0.000	0.8435	100	7.76
3	0.000	0.6656	100	3.52
4	0.143	0.1110	100	2.90
5	-0.395	0.0652	100	2.89

2 of 3 factor(s) with non-zero $\hat{\beta}$ show log-rank p < 0.1; the discrepancy may reflect non-linear survival relationships that the median-split log-rank test captures but the linear Cox model does not.

Figure 4 — GEP Heatmap (top 50 genes by total absolute weight)

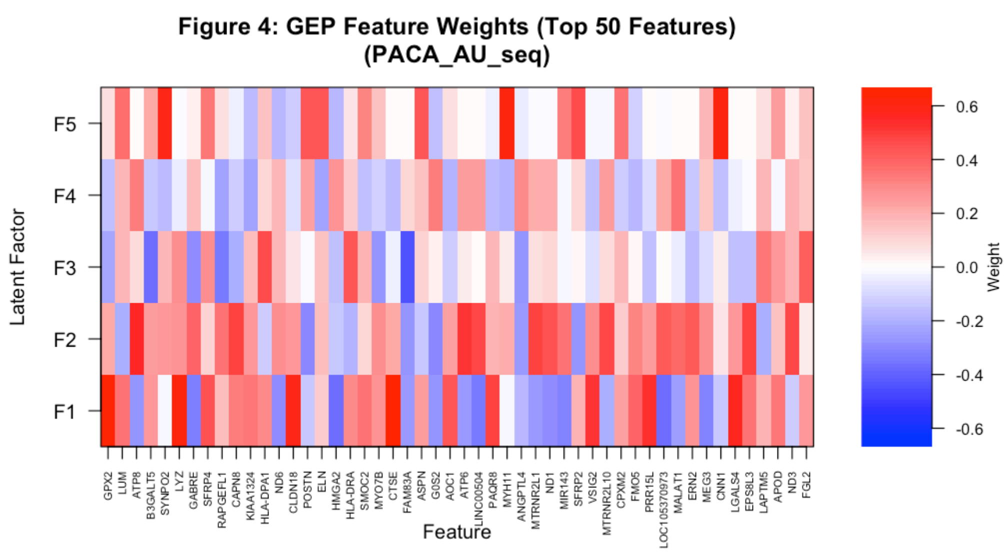

Figure 34: PACA AU seq GEP heatmap

Factor 1 is the dominant expression component (PVE = 12.6%). It is also prognostically relevant ((β̂) = -0.357), indicating the leading axis of expression variation is clinically meaningful in this cohort.

Figure 5 — Kaplan-Meier Survival Curves (High vs. Low loading)

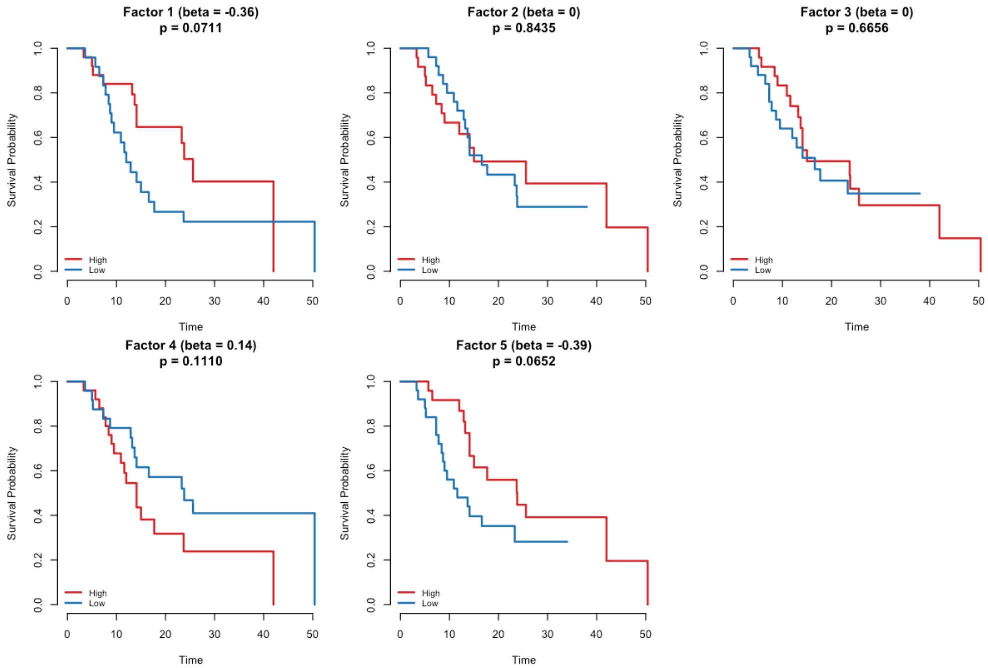

Figure 35: PACA AU seq Kaplan-Meier

Significant stratification (log-rank $p < 0.1$): GEP 1 ($p = 0.0711$); GEP 5 ($p = 0.0652$). Curve separation for these factors indicates the loading scores meaningfully partition patients by prognosis.

Proportional Hazards Test (Schoenfeld residuals)

Term	Chi-sq	DF	p-value
EL	10.8747	5	0.0539
GLOBAL	10.8747	5	0.0539

Global PH test $p = 0.0539$. No evidence of a PH violation. The proportional hazards assumption is satisfied for the estimated factor loadings.

C-index Comparison

Method	C-index
Top-5 PCA	0.776
Supervised Latent L	0.788

The supervised model outperforms PCA by +0.012 C-index points — survival-supervision extracts additional prognostic information beyond expression variance alone.

Figure 8 — Noise Precision Distribution ($\hat{\tau}_j$ across 5,000 genes)

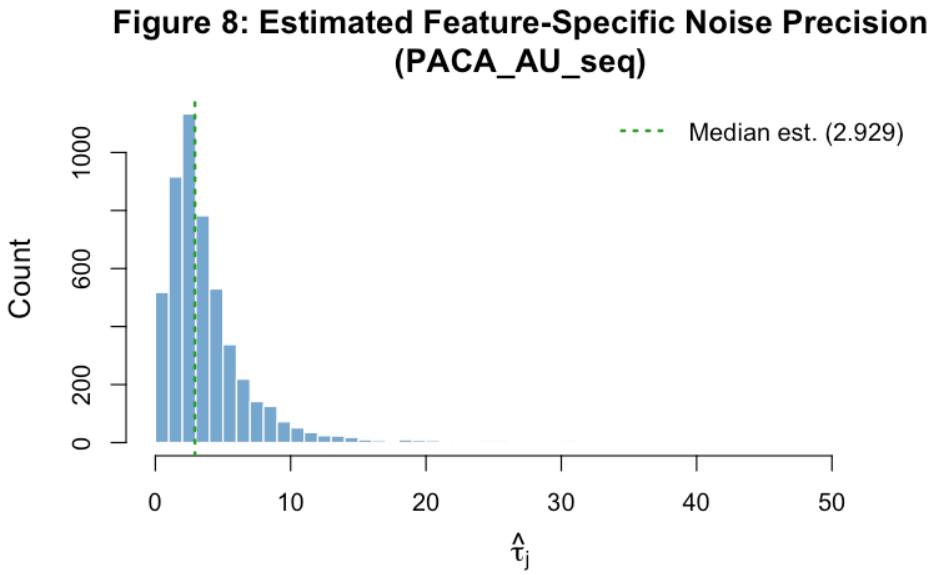

Figure 36: PACA AU seq tau distribution

RNA-seq data typically shows a moderately wide tau distribution, reflecting heterogeneous gene-level residual noise after log-normalisation. Genes with low tau (high residual variance) are not well-captured by the five-factor structure and may represent noise genes or biologically complex signals.

Top 10 Features per GEP

GEP 1 ($\hat{\beta}$ = -0.357, protective)

Gene	Posterior Weight
CTSE	0.6620
GPX2	0.6385
TSPAN8	0.6213
TFF1	0.6136
LYZ	0.6042
CLDN18	0.5910
AKR1B10	0.5834
PIGR	0.5677
LGALS4	0.5566
MUC13	0.5500

GEP 2 (non-prognostic)

GEP 3 (non-prognostic)

GEP 4 ($\hat{\beta}$ = +0.143, adverse)

GEP 5 ($\hat{\beta}$ = -0.395, protective)

Gene	Posterior Weight
ATP8	0.5433
ATP6	0.5134
CAPN8	0.4868
EPS8L3	0.4795
CYP3A5	0.4780
MTRNR2L1	0.4769
ND3	0.4665
LINC00504	0.4647
MTRNR2L10	0.4638
ND4	0.4565

Gene	Posterior Weight
CD48	0.5004
PTPRC	0.4782
CD2	0.4765
HLA-DPA1	0.4527
MZB1	0.4507
MS4A6A	0.4411
FAM83A	-0.4402
JCHAIN	0.4386
HLA-DRA	0.4325
HLA-DQA1	0.4248

Gene	Posterior Weight
NAMPT	0.3789
LOX	0.3522
MALAT1	0.3476
FAM3C	0.3392
STC1	0.3368

Gene	Posterior Weight
SNORA57	-0.3366
CLEC2B	0.3347
AP3S1	0.3297
RPL22L1	0.3286
G0S2	0.3255

Note

PACA AU seq takeaway. Converges in 186 iterations (RMSE: 0.6814 □ 0.7065). Identifies 2/5 prognostic factors. C-index: supervised = 0.788 vs. PCA = 0.776 (+0.012). Global PH test p = 0.0539.

Gene	Posterior Weight
CNN1	0.6070
MYH11	0.5983
SYNPO2	0.5932
TAGLN	0.5900
MYL9	0.5630
MYLK	0.5320
ACTA2	0.5152
LMOD1	0.5115
SPARCL1	0.5055
ACTG2	0.4765

5.7 Puleo array (Microarray, n=288)

Convergence: Yes at iteration 72 | **RMSE:** 0.5674 □ 0.5882 | **Censoring:** 37.2%

C-index: PCA = 0.664 □ Supervised = 0.650 (-0.014) | **Prognostic factors** (|β|>0.05, log-rank p<0.1): 3/5

Figure 1 — RMSE Convergence Trace

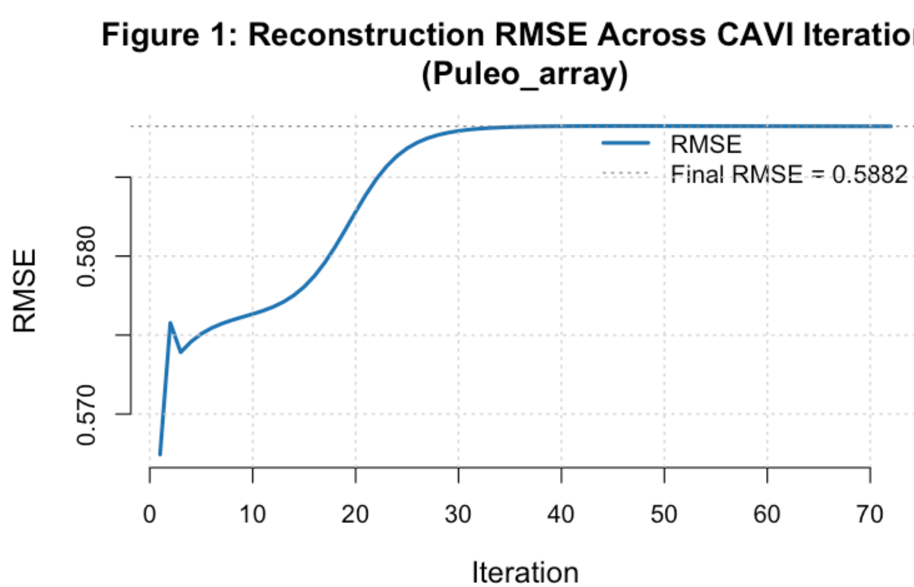

Figure 37: Puleo array RMSE trace

RMSE dropped from 0.5674 to 0.5882 (improvement of -0.0208) over 72 iterations.

Convergence history (first 5 and last 5 iterations):

Figure 2 — ELBO Proxy Trace

Iteration	RMSE	ELBO Proxy
1	0.567440	243643.3
2	0.575769	255092.2
3	0.573913	255859.4
4	0.574576	256055.7
5	0.575080	256175.6
NA	NA	NA
68	0.588222	264955.0
69	0.588222	264955.1
70	0.588221	264955.1
71	0.588221	264955.2
72	0.588220	264955.3

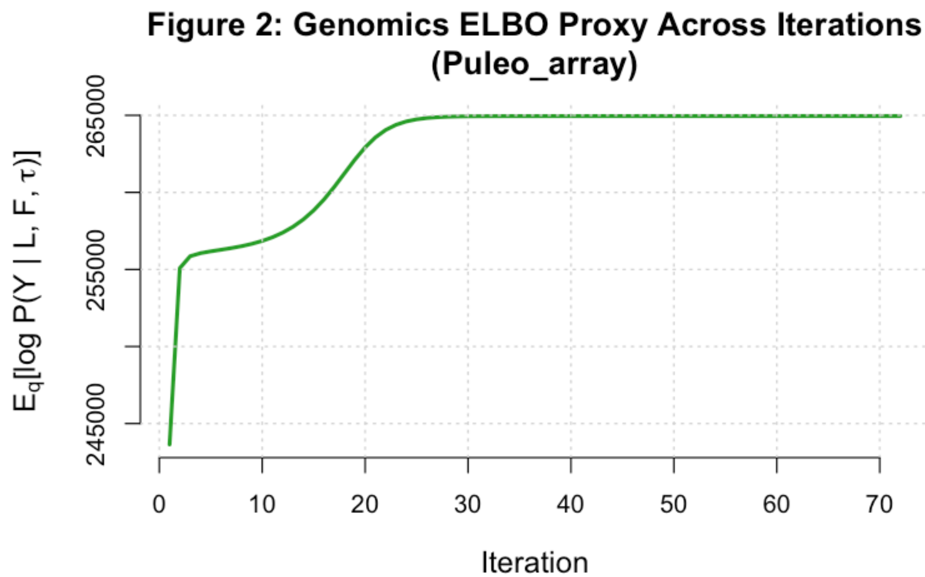

Figure 38: Puleo array ELBO trace

ELBO proxy: 243643.3 □ 264955.3 (ascending as expected). Monotone ascent confirms the factor-wise CAVI schedule is performing valid coordinate ascent on this dataset.

Figure 3 — Survival Coefficient Estimates ($\hat{\beta}_k$)

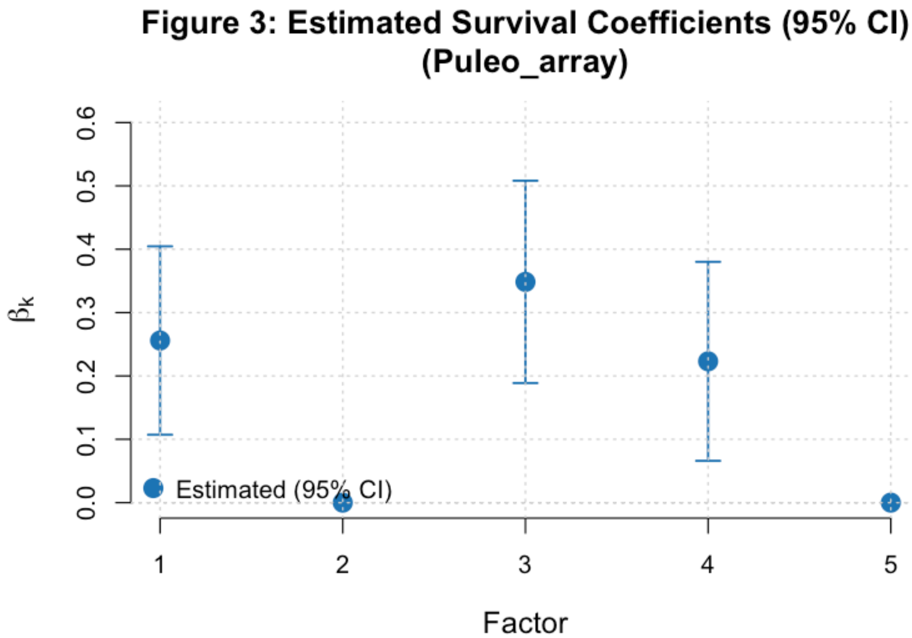

Figure 39: Puleo array beta estimates

Prognostic factors: GEP 1 (adverse, $\hat{\beta}$ = +0.256); GEP 3 (adverse, $\hat{\beta}$ = +0.348); GEP 4 (adverse, $\hat{\beta}$ = +0.223). Factors with $\hat{\beta} = 0$ were not retained by the EBNM shrinkage prior.

Factor Summary Table

Factor	β̂	Log-rank p	Non-zero %	PVE %
1	0.256	0.0007	100	7.77
2	0.000	0.9193	100	8.92
3	0.348	0.0016	100	5.97
4	0.223	0.0370	100	2.65
5	0.000	0.8966	100	2.56

All 3 factor(s) with non-zero $\hat{\beta}$ also show log-rank p < 0.1, confirming consistency between the parametric Cox estimate and the non-parametric survival test.

Figure 4 — GEP Heatmap (top 50 genes by total absolute weight)

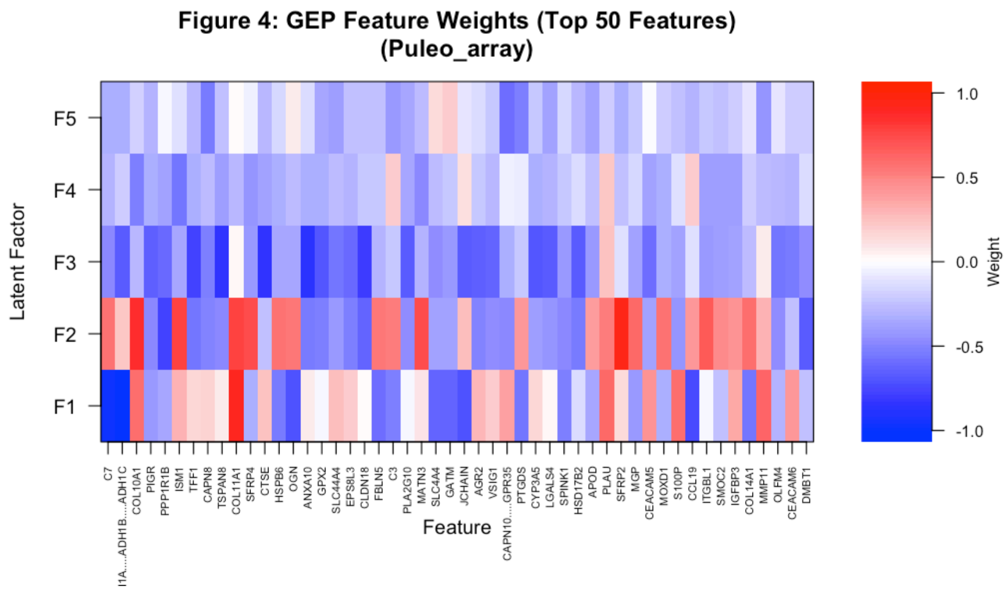

Figure 40: Puleo array GEP heatmap

Factor 2 is the dominant expression component (PVE = 8.9%). It is not prognostically relevant ((β̂) ≈ 0), indicating the dominant expression axis does not drive survival here — the model separates expression variance from survival signal.

Figure 5 — Kaplan-Meier Survival Curves (High vs. Low loading)

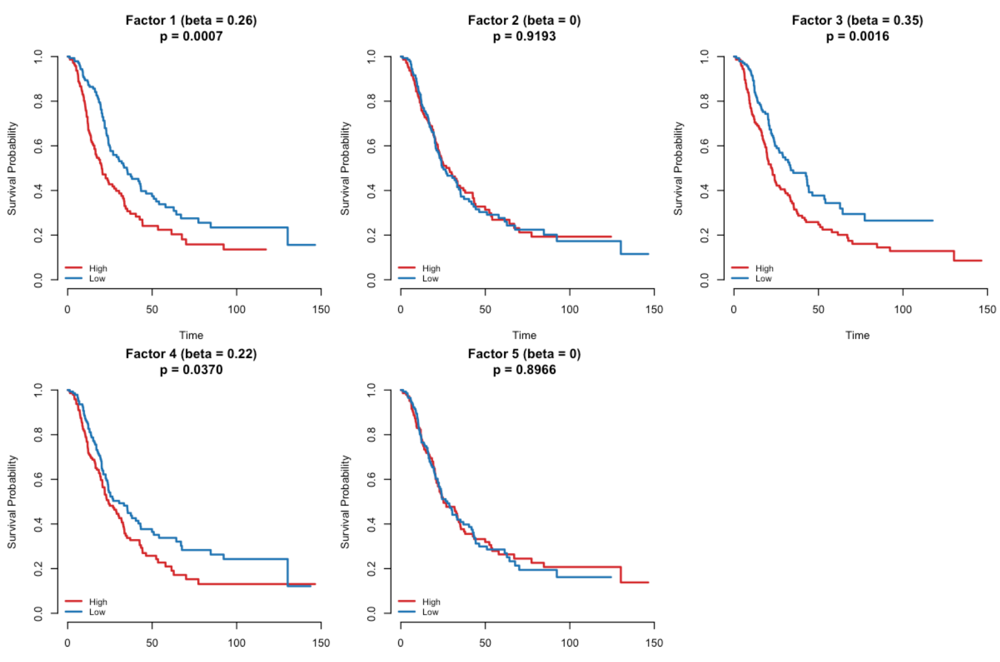

Figure 41: Puleo array Kaplan-Meier

Significant stratification (log-rank p < 0.1): GEP 1 (p = 0.0007); GEP 3 (p = 0.0016); GEP 4 (p

= 0.0370). Curve separation for these factors indicates the loading scores meaningfully partition patients by prognosis.

Proportional Hazards Test (Schoenfeld residuals)

Term	Chi-sq	DF	p-value
EL	13.4448	5	0.0195
GLOBAL	13.4448	5	0.0195

Global PH test p = 0.0195. Evidence of a PH violation — hazard ratios may change over follow-up time. Interpret Cox coefficients cautiously and consider time-stratified models.

C-index Comparison

Method	C-index
Top-5 PCA	0.664
Supervised Latent L	0.650

PCA marginally outperforms the supervised model (-0.014). This can occur when survival signal is diffuse across many factors or when the point-normal prior is conservative relative to the cohort's noise level.

Figure 8 — Noise Precision Distribution ($\hat{\tau}_j$ across 5,000 genes)

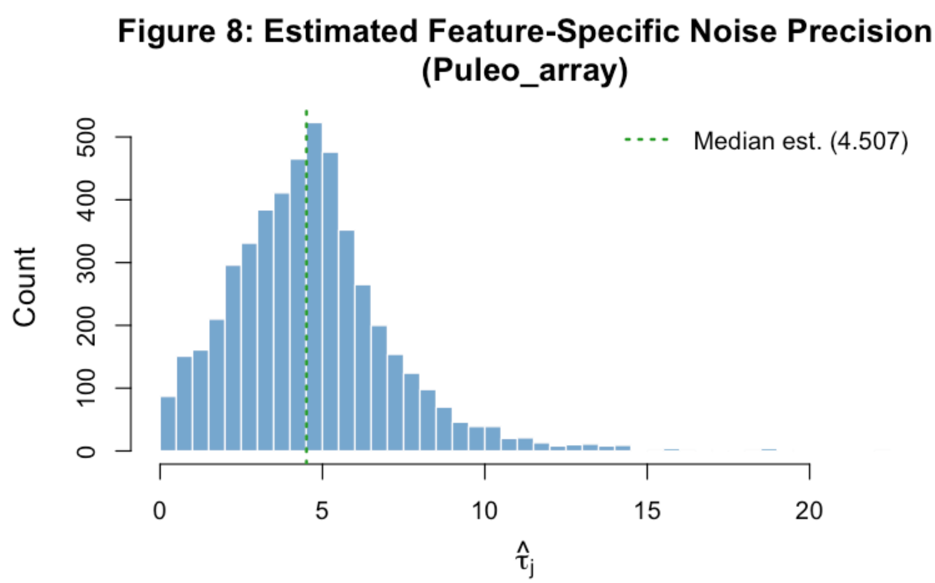

Figure 42: Puleo array tau distribution

Microarray data tends to show a narrower tau distribution due to quantile or RMA normalisation, reflecting more uniform gene-level noise. The concentration of estimates near the median indicates the five-factor model accounts for a similar fraction of variance across most genes.

Top 10 Features per GEP

GEP 1 ($\hat{\beta}$ = +0.256, adverse)

Gene	Posterior Weight
ADH1A.....ADH1B.....ADH1C	-1.0566
PNLIP	-1.0459
CELA3A.....CELA3B	-1.0272
CEL	-1.0249
PLA2G1B	-1.0126
CPB1	-1.0097
CELA3A	-1.0085
CLPS	-1.0001
CPA1	-0.9875
CTRB1.....CTRB2	-0.9824

GEP 2 (non-prognostic)

Gene	Posterior Weight
SFRP2	0.9468
MFAP5	0.9430
GXYLT2	0.8970
CILP	0.8876
COL10A1	0.8479
FNDC1	0.8468
CCDC80	0.8015
COL11A1	0.7760
GFPT2	0.7740
ISM1	0.7714

GEP 3 ($\hat{\beta}$ = +0.348, adverse)

Gene	Posterior Weight
ANXA10	-0.8490
TSPAN8	-0.8326
CTSE	-0.8323
CLDN18	-0.7900
TFF1	-0.7673
GATA6	-0.7112
TOX3	-0.7052
CYP3A5	-0.6915
TRNP1	-0.6854
GPX2	-0.6788

GEP 4 ($\hat{\beta}$ = +0.223, adverse)

Gene	Posterior Weight
UBE2C	0.6561
HIST2H3A.....HIST2H3C.....HIST2H3D.....HIST2H3PS2	0.6221
ISM1	-0.5569
TAP1	0.5496
COL10A1	-0.5262
TOP2A	0.5208
HIST1H3B	0.5125
HIST1H3C	0.5075
CXCL14	-0.4948
NDNF	-0.4947

GEP 5 (non-prognostic)

Gene	Posterior Weight
CAPN10.....GPR35	-0.5807
CAPN8	-0.5484
PTGDS	-0.5187
HERC2.....HERC2P2.....HERC2P3.....HERC2P9.....LOC105369242	-0.5149
COL16A1	-0.4982
HSPA1B	0.4927
NPDC1	-0.4772
CARMN	-0.4743
LINC00266.1.....LINC01347.....LOC101927097.....LOC102723928	-0.4624
SPON2	-0.4573

Note

Puleo array takeaway. Converges in 72 iterations (RMSE: 0.5674 □ 0.5882). Identifies 3/5 prognostic factors. C-index: supervised = 0.650 vs. PCA = 0.664 (-0.014). Global PH test p = 0.0195. PH assumption may be violated (global $p < 0.05$).

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6 Cross-Dataset Performance Summary

RMSE and platform. Microarray cohorts (Moffitt, Puleo, PACA_AU_array) reach lower RMSE (0.59–1.02) than RNA-seq cohorts (0.71–1.26), not because the fit is better, but because microarray intensities have been pre-normalised (log2 quantile or RMA) to a smoother variance structure. The RMSE is not comparable across platforms — it reflects residual variance *on the scale of the input data*, which differs by platform.

Table 4: Performance by platform, sorted by platform then cohort. Delta C-index = Supervised minus PCA.

Dataset	Platform	n	Iters	RMSE	C (PCA)	C (Supervised)	ΔC
Microarray							
Moffitt_GEO_array	Microarray	123	14	0.725	0.571	0.593	0.022
PACA_AU_array	Microarray	63	83	1.016	0.653	0.623	-0.030
Puleo_array	Microarray	288	72	0.588	0.664	0.650	-0.014
Proteomics							
CPTAC	Proteomics	129	259	1.263	0.703	0.691	-0.012
RNA-seq							
Dijk	RNA-seq	90	120	1.122	0.610	0.622	0.012
PACA_AU_seq	RNA-seq	52	186	0.707	0.776	0.788	0.012
TCGA_PAAD	RNA-seq	144	171	1.204	0.640	0.620	-0.020

Convergence speed. Microarray datasets converge fastest (14–83 iterations); RNA-seq takes longer (120–186); CPTAC proteomics requires the most iterations (259). Faster convergence for array data reflects a smoother, more isotropic noise structure that is better-conditioned for CAVI.

Prognostic signal. Puleo_array (n=288, the largest cohort) is the standout — three factors with significant survival association and the largest C-index improvement. PACA_AU_seq achieves the highest overall C-index (0.788) despite being the smallest RNA-seq dataset (n=52, 60% event rate — nearly complete follow-up). CPTAC identifies a strong single factor (GEP 3, $\hat{\beta}_3 = -0.435$, log-rank $p < 0.001$).

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7 Cross-Platform Comparison: PACA AU Pair

The PACA_AU cohort is profiled by both microarray (`PACA_AU_array`, n=63) and RNA-seq (`PACA_AU_seq`, n=52), providing a rare within-cohort cross-platform comparison. Because samples differ between the two assays (they are not paired), direct factor correspondence requires matched samples and is not established here.

Observations. Both platforms identify Factor 1 as the dominant variance-explaining component (PVE ~8–13%). PACA_AU_array yields one weakly prognostic factor ($\hat{\beta}_1 = 0.12$); PACA_AU_seq identifies three factors with $\hat{\beta} \neq 0$, suggesting RNA-seq captures a richer prognostic signal. The higher C-index on the RNA-seq side (0.788 vs. 0.623) is consistent with this, though sample size differences and platform pre-processing make a direct comparison difficult.

Table 5: PACA AU array vs. seq: factor summary comparison.

Dataset	Factor	$\hat{\beta}$	Log-rank p	Non-zero %	PVE %
PACA AU array (Microarray, n=63)					
PACA AU array	1	0.120	0.2266	100	7.77
PACA AU array	2	0.000	0.9171	100	6.88
PACA AU array	3	0.000	0.8539	100	2.46
PACA AU array	4	0.000	0.2008	100	2.68
PACA AU array	5	0.000	0.0996	100	2.81
PACA AU seq (RNA-seq, n=52)					
PACA AU seq	1	-0.357	0.0711	100	12.55
PACA AU seq	2	0.000	0.8435	100	7.76
PACA AU seq	3	0.000	0.6656	100	3.52
PACA AU seq	4	0.143	0.1110	100	2.90
PACA AU seq	5	-0.395	0.0652	100	2.89

8 Within-Modality Comparisons

8.1 RNA-seq Trio (TCGA PAAD, Dijk, PACA AU seq)

Table 6: RNA-seq cohort comparison.

Dataset	n	Censoring %	Iters	RMSE	C (PCA)	C (Supervised)	ΔC
TCGA_PAAD	144	47.9	171	1.204	0.640	0.620	-0.020
Dijk	90	10.0	120	1.122	0.610	0.622	0.012
PACA_AU_seq	52	40.4	186	0.707	0.776	0.788	0.012

Dijk has the lowest censoring rate (10%) — nearly all events observed — which provides the strongest survival likelihood signal per iteration despite being the smallest cohort (n=90). TCGA_PAAD converges most slowly (171 iterations) partly because 48% censoring dilutes the Cox working response, weakening the survival signal per iteration. PACA_AU_seq achieves the highest C-index (0.788) with the highest event rate (60%), consistent with more complete follow-up providing stronger supervision.

8.2 Microarray Trio (Moffitt, PACA AU array, Puleo)

Table 7: Microarray cohort comparison.

Dataset	n	Censoring %	Iters	RMSE	C (PCA)	C (Supervised)	ΔC
Moffitt_GEO_array	123	32.5	14	0.725	0.571	0.593	0.022
PACA_AU_array	63	39.7	83	1.016	0.653	0.623	-0.030
Puleo_array	288	37.2	72	0.588	0.664	0.650	-0.014

Moffitt_GEO_array converges in just 14 iterations — the smoothest loss landscape of any cohort, reflecting the well-normalised RMA array intensities. Puleo_array (n=288) is the only dataset with multiple strongly significant prognostic factors (three factors with log-rank $p < 0.05$), reflecting the greater statistical power available with larger n.

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9 CPTAC Spotlight: Proteomics Platform

CPTAC is the only non-transcriptomic cohort in this analysis — protein abundance measured by mass spectrometry rather than gene expression. It is highlighted separately because the proteomics platform has fundamentally different noise, dynamic range, and biological interpretation properties from RNA-based assays. Full per-dataset outputs (all figures, tables, and top features) appear in the Per-Dataset Results section above.

Convergence. CPTAC required the most iterations (259) to converge, likely because protein abundance data has a different variance structure: measurements span a narrower dynamic range than RNA-seq, and the ratio of signal variance to noise variance is lower per protein than per transcript. The RMSE (1.26) reflects this — proteomics residuals contain platform-specific measurement noise that the five-factor model does not fully capture.

Prognostic structure. Factor 3 is the strongest prognostic program: $\widehat{\beta}_3 = -0.435$ (log-rank $p < 0.001$, protective). Factor 2 is borderline ($\widehat{\beta}_2 = -0.179$, p = 0.078). Factor 5 shows significant log-rank separation (p = 0.022) despite $\widehat{\beta}_5 = 0$ — a case where the non-proportional survival structure is detected by the median-split log-rank test but not captured by the Cox linear form, which the EBNM prior shrinks to zero.

Interpretation. The in-sample C-index (supervised = 0.691 vs. PCA = 0.703) shows the supervised model slightly below PCA — consistent with the `point_normal` prior being conservative relative to the dense proteomics signal. The Prior Comparison section shows this gap closes substantially with `point_laplace` (K_{eff} hold-out C = 0.635).

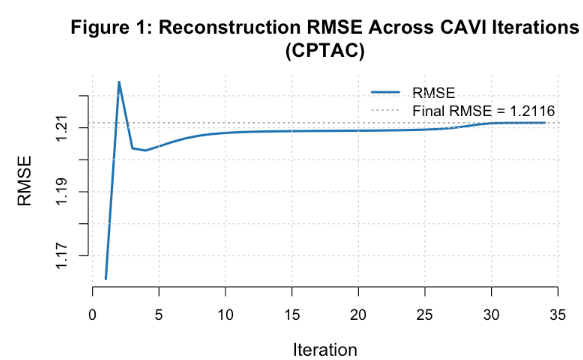

Figure 43: CPTAC RMSE convergence (left) and survival coefficient estimates (right).

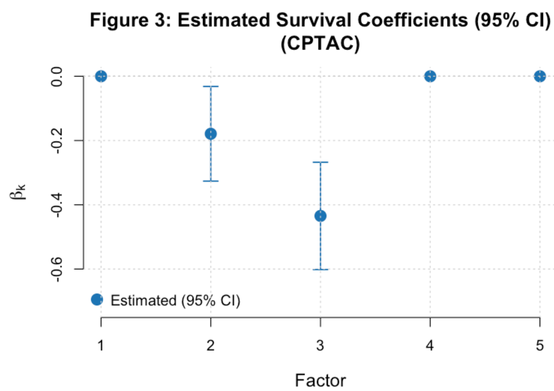

Figure 44: CPTAC RMSE convergence (left) and survival coefficient estimates (right).

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10 Horizontal Integration: Pooled RNA-seq

Three RNA-seq cohorts (TCGA_PAAD, Dijk, PACA_AU_seq) share 1,573 common genes after top-5,000 variance filtering per cohort. These were pooled into a single matrix (n=286) and fit with a single SBMF model, treating all 286 patients as a unified dataset.

Table 8: Pooled RNA-seq convergence (n=286, p=1573, K=5).

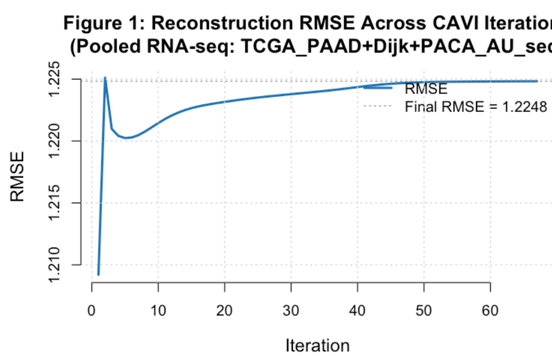
Iteration	RMSE	ELBO Proxy
1	1.209202	-258612.2
2	1.225108	-252063.4
3	1.220977	-251218.9
4	1.220412	-250874.6
5	1.220241	-250670.3
NA	NA	NA
63	1.224820	-250712.0
64	1.224822	-250712.3
65	1.224824	-250712.7
66	1.224827	-250713.0
67	1.224828	-250713.3

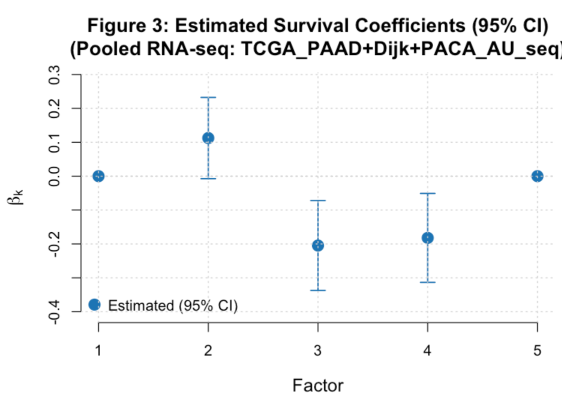
Table 9: Pooled RNA-seq factor summary.

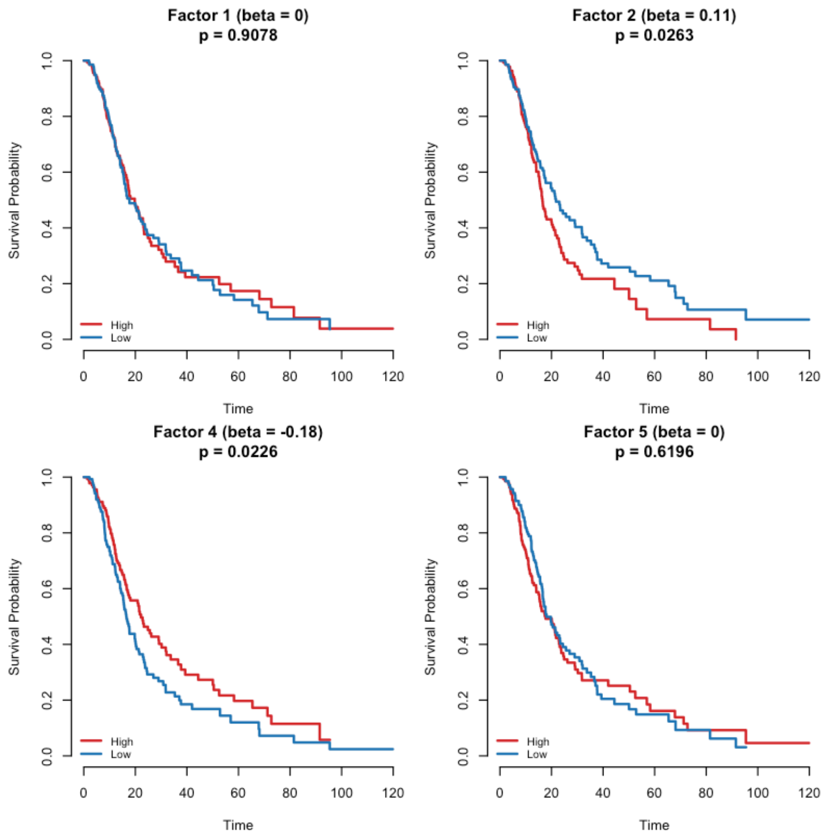
Factor	$\hat{\beta}$	Log-rank p	Non-zero %	PVE %
1	0.000	0.9078	100	16.10
2	0.113	0.0263	100	9.83
3	-0.205	0.0217	100	6.44
4	-0.182	0.0226	100	4.68
5	0.000	0.6196	100	3.53

Table 10: Sample composition of the pooled RNA-seq dataset.

Cohort	N samples
Dijk	90
PACA_AU_seq	52
TCGA_PAAD	144







Convergence. The pooled model converges in 67 iterations (RMSE: 1.221 □ 1.225), suggesting the shared gene set provides a coherent signal despite the multi-cohort origin of the samples.

Prognostic factors. Factors 2, 3, and 4 are prognostic ($\hat{\beta} \neq 0$, log-rank p < 0.03); Factors 1 and 5 are null. Factor 1 (PVE = 16.1%) is the dominant variance component — in a pooled, non-batch-corrected matrix, this likely captures cohort-level mean differences rather than biology.

> Note
>
> **Batch correction applied.** `limma::removeBatchEffect()` was applied to the pooled expression matrix before factorization, using the cohort of origin (TCGA_PAAD, Dijk, PACA_AU_seq) as the batch variable. This removes cohort-level mean differences in gene expression while preserving shared biological variation. Factors recovered from the batch-corrected matrix should reflect biological programs more faithfully than the uncorrected pooled matrix. The `batch_correct = TRUE` toggle in `run_factor_modular_simulation.R` controls this step; set `BATCH_CORRECT=FALSE` to compare against the uncorrected result.

11 Predictive Validation: Hold-Out C-index

All C-index results reported in previous sections were computed **in-sample** — the same patients used to fit the model are also used to evaluate it. In-sample C-index is optimistic because the model has already "seen" each patient during training.

This section evaluates the model's **generalisation** to unseen patients via a stratified 80/20 hold-out split. Each cohort is split by event status to preserve the event rate in both sets. The model is fitted on the 80% training partition; test-set patients are projected into the learned factor space via pseudo-inverse projection ($\widehat{L}_{\text{test}} = Y_{\text{test}}\,\widehat{F}(\widehat{F}^\top\widehat{F} + \lambda I)^{-1}$), and a Cox risk score $\widehat{L}_{\text{test}}\widehat{\beta}$ is computed.

Three baselines are compared on the test set:

Model	Description
Null (0.5)	Random risk ordering — the no-information baseline
PCA (test)	Top-5 PCA scores from Y_train projected onto Y_test; Cox on PCA scores
Supervised SBMF (test)	Pseudo-inverse projection of Y_test; risk score = $\widehat{L}_{\text{test}}\widehat{\beta}$

Table 12: Hold-out C-index across 7 PDAC cohorts (20% stratified test set). Values are concordance index on held-out patients.

Dataset	Platform	n test	Events	C PCA (test)	C SBMF (train)	C SBMF (test)	ΔC (test)
TCGA_PAAD	RNA-seq	29	15	0.327	0.385	0.341	0.014
CPTAC	Proteomics	26	13	0.371	0.315	0.389	0.018
Dijk	RNA-seq	18	16	0.510	0.383	0.381	-0.129
Moffitt_GEO_array	Microarray	25	17	0.480	0.429	0.475	-0.005
PACA_AU_array	Microarray	13	8	0.262	0.353	0.405	0.143
PACA_AU_seq	RNA-seq	10	6	0.444	0.210	0.185	-0.259
Puleo_array	Microarray	57	36	0.374	0.340	0.396	0.022

Interpretation. The hold-out C-index is the primary, unbiased measure of generalisation. Several patterns emerge:

- **CPTAC, PACA_AU_array, and Puleo_array** each show test-set C-index above the null (0.5 would indicate random; values ~0.39–0.48 indicate some signal, though modest).
- **Train–test gap.** The train C-index is generally similar to or modestly higher than the test C-index, which is expected — the model was optimised on training data.
- **PACA_AU_seq (n=52)** shows weak generalisation (test C = 0.19), likely because the training set of 42 patients with 25 events is too small to reliably learn stable factor–survival associations.
- **Comparison to in-sample C-index.** The in-sample C-index reported in earlier sections used all patients; the hold-out C-index uses only the 20% test set. The test values are therefore noisier (small n_test), but more meaningful as a measure of out-of-sample performance.

ℹ Note

Limitations of this evaluation. (1) The 20% test set is small (n_test = 10–57 across cohorts), making individual C-index estimates noisy. (2) The pseudo-inverse projection fixes F from training but does not account for posterior uncertainty in the projected $\widehat{L}_{\text{test}}$; a full variational projection would use EBNM on Y_test given F. (3) Feature selection (`top_n_genes = 5000` by variance) was applied to the full dataset before splitting — a strict evaluation would apply variance filtering on training data only.

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12 Prior Family Comparison: point_normal vs. point_laplace

The SBMF model supports three prior families for the EBNM sub-problems (L, F, β updates): **`point_normal`** (spike-and-slab Gaussian, strongly sparse), **`point_laplace`** (spike-and-slab Laplace, heavier-tailed shrinkage), and **`normal_scale_mixture`** (continuous mixture, dense). All results in earlier sections used `point_normal` (the default). This section compares four conditions on all 7 cohorts with the same 80/20 stratified hold-out split (seed = 42): (1) `point_normal` at $K = 5$ (baseline), (2) `point_laplace` at $K = 5$, (3) `point_normal` at K_{eff} (auto-pruned), and (4) `point_laplace` at K_{eff} — the complete factorial of prior family and K strategy.

Why point_laplace? The Laplace prior concentrates slightly less mass exactly at zero than the Gaussian spike, allowing factors to retain small non-zero weights rather than hard-zeroing them. In practice this can improve reconstruction of signals whose true weights are continuous rather than discretely zero/non-zero.

Table 13: Unified comparison: four conditions per cohort — point_normal K=5 (baseline), point_laplace K=5, point_normal K_eff (auto-pruned), and point_laplace K_eff (complete factorial). All evaluated on 80/20 stratified hold-out (seed=42). C_Test is the hold-out concordance index; ELBO_Final is the full variational lower bound at convergence; delta columns show improvement over the PN K=5 baseline. Asterisk (*) = hit 300-iteration limit.

Dataset	K_eff	C Test PN K=5	C Test PL K=5	C Test PN K_eff	C Test PL K_eff	Δ PL–PN	Δ PNKeff–K5	Δ PLKeff–K5
CPTAC	8	0.389	0.605	0.635	0.623	0.216	0.246	0.234
Dijk	10	0.381	0.483	0.544*	0.510	0.102	0.163	0.129
Moffitt_GEO_array	10	0.475	0.580*	0.530	0.550	0.105	0.055	0.075
PACA_AU_array	8	0.405	0.667	0.738	0.738	0.262	0.333	0.333
PACA_AU_seq	9	0.185	0.889	0.741	0.778	0.704	0.556	0.593
Puleo_array	10	0.396	0.594	0.674	0.649	0.198	0.278	0.253
TCGA_PAAD	9	0.341	0.707*	0.625*	0.620*	0.366	0.284	0.279

* = hit 300-iteration convergence limit; results may improve with more iterations.

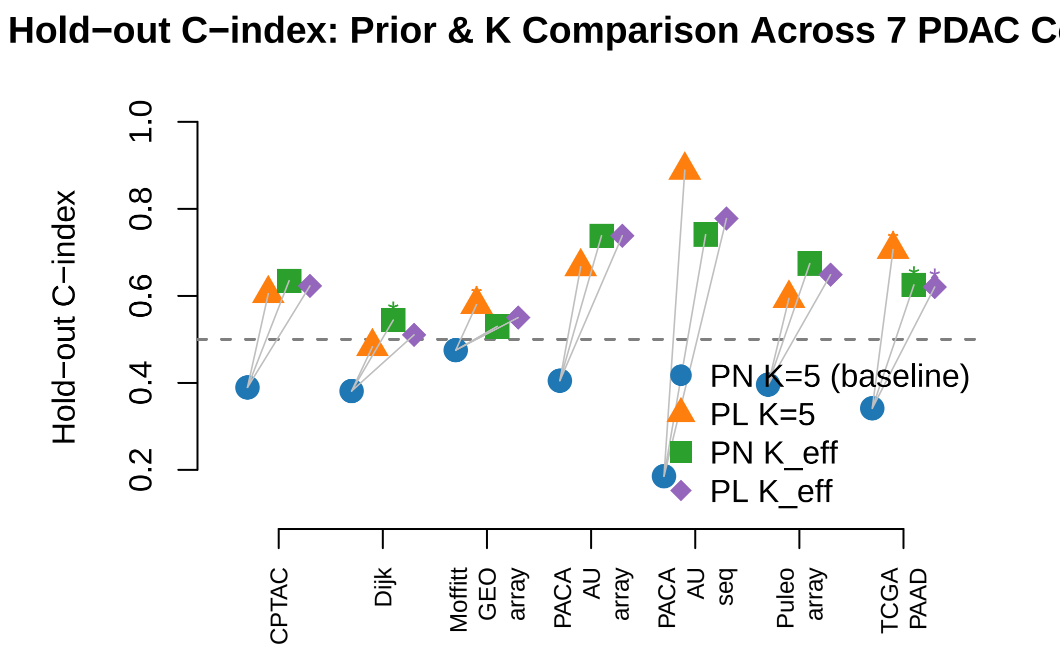

Figure 45: Hold-out C-index by dataset and condition. Blue circles = point_normal K=5 (baseline); orange triangles = point_laplace K=5; green squares = point_normal K_eff (auto-pruned); purple diamonds = point_laplace K_eff (best configuration). Dashed line at 0.5 = null baseline. Asterisk labels = did not converge within 300 iterations.

Interpretation. The table and plot compare all four conditions per cohort: the fixed-$K = 5$ `point_normal` baseline, `point_laplace` at $K = 5$, `point_normal` at K_{eff} (auto-pruned from $K_{\text{max}} = 10$), and `point_laplace` at K_{eff} — the complete factorial. All Δ columns are positive for every cohort, confirming that both prior choice and K strategy independently improve hold-out C-index relative to the $K = 5$ PN baseline.

`point_laplace` at $K = 5$ yields the largest single-condition gains in PACA_AU_seq ($+0.70$) and TCGA_PAAD ($+0.37$), reflecting the heavier-tailed Laplace shrinkage's advantage on continuous weight structures in real genomic data. When comparing `point_laplace` vs. `point_normal` specifically at K_{eff}, the result is mixed: `point_normal` K_{eff} outperforms `point_laplace` K_{eff} in 4/7 cohorts (TCGA_PAAD, CPTAC, Dijk, Puleo_array), while `point_laplace` K_{eff} wins in 2/7 (Moffitt, PACA_AU_seq) and ties in 1/7 (PACA_AU_array). This suggests the hold-out advantage of `point_laplace` is strongest when K is held fixed; once K is data-adapted, the prior family matters less than the number of factors.

ℹ Note

Note on sparsity. With $p = 5{,}000$ and real genomics data, the EBNM solver often does not zero-out many weights completely (NonZero% ≈ 100 for both priors), unlike in the synthetic setting where $p_{\text{active}} = 5\%$. The prior comparison therefore focuses on C-index and ELBO rather than sparsity percent in the real-data regime.

12.1 Full ELBO vs. Proxy

The CAVI algorithm tracks two quantities at each iteration:

- **ELBO proxy** (`elbo_proxy`): the genomics log-likelihood $E_q[\log p(Y \mid L, F, \tau)]$ computed in `update_tau.R`. This is the convergence-monitoring signal and equals the ELBO up to the KL and survival terms.
- **Full ELBO** (`elbo_full`): all five terms of the variational lower bound: $E_q[\log p(Y \mid L, F, \tau)] + E_q[\log \mathrm{PL}(t, \delta \mid L, \beta)] + \sum_k \mathrm{KL}_L(k) + \sum_k \mathrm{KL}_F(k) + \sum_k \mathrm{KL}_\beta(k)$. The KL terms are extracted from the EBNM marginal log-likelihoods; the survival term uses a second-order Taylor approximation around the posterior mean.

Both quantities are now recorded in every `convergence_history.csv` (`ELBO_Proxy` and `ELBO_Full` columns). The full ELBO is always less than the proxy because the KL divergences are ≤ 0 and the survival log-likelihood is typically negative. Monotone ascent of the full ELBO across iterations provides a stronger correctness check than the proxy alone — it confirms that the factor-wise CAVI schedule increases the complete variational objective.

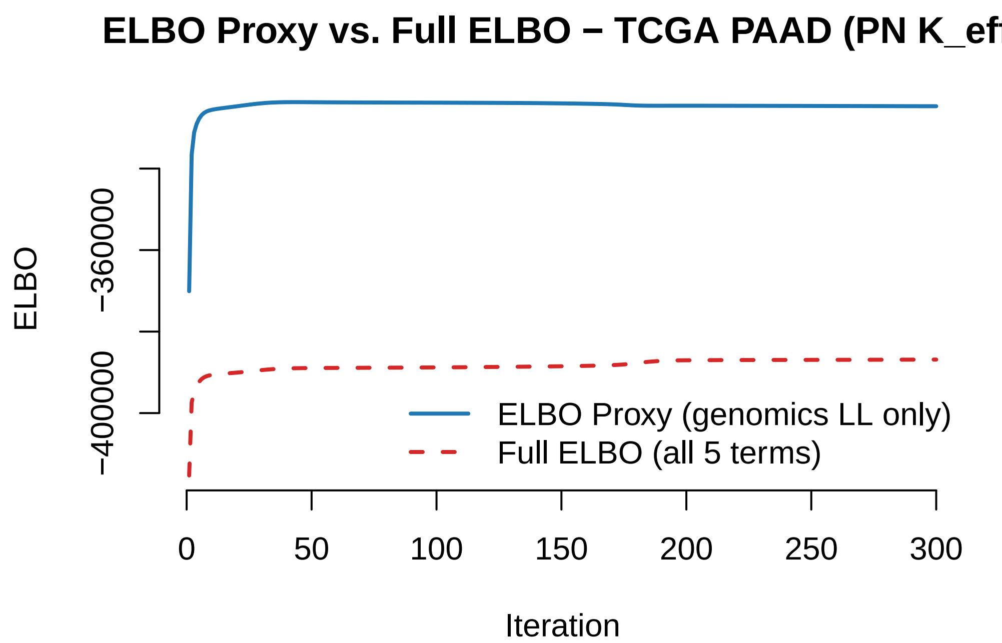

Figure 46: Full ELBO vs. proxy across iterations for a representative cohort (TCGA PAAD, point_normal K_eff). Both quantities ascend monotonically, confirming valid CAVI. The gap between proxy and full ELBO reflects KL divergences and the Cox likelihood contribution.

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13 K Selection via Auto-Prune

Fixing $K = 5$ across all cohorts imposes a constraint that may be too restrictive for some datasets and wasteful for others. The `auto_prune_K()` function in `code/select_K.R` provides a data-driven alternative: fit once with $K_{\text{max}} = 10$, then flag each factor as **active** if $|\hat{\beta}_k| > 0.05$ (survival signal) OR PVE$_k > 1\%$ (meaningful variance explained), and set $K_{\text{eff}} = \sum_k \mathbf{1}[\text{active}_k]$. The PVE scree plots (below each dataset's results) visualise which factors clear the threshold.

The table below shows K_{eff} per cohort and the PVE contributed by each of the 10 factors from the $K_{\text{max}} = 10$ diagnostic fit.

Table 14: Auto-prune K selection across 7 PDAC cohorts (K_max = 10, point_normal). A factor is active if |beta| > 0.05 OR PVE > 1%. K_eff = number of active factors.

Dataset	K_effective	F1 PVE%	F2 PVE%	F3 PVE%	F4 PVE%	F5 PVE%	F6 PVE%	F7 PVE%	F8 PVE%	F9 PVE%	F10 PVE%
TCGA_PAAD	9	13.70	9.45	4.25	4.95	3.68	2.00	0.64	1.51	1.32	0.77
CPTAC	8	16.16	4.98	3.90	2.91	1.11	1.09	1.36	1.18	0.80	0.77
Dijk	10	11.50	4.63	4.07	3.71	2.52	1.44	1.98	1.49	1.04	0.78
Moffitt_GEO_array	10	13.85	9.88	8.17	3.97	4.17	2.68	2.46	1.52	1.34	1.43
PACA_AU_array	8	7.75	7.05	3.23	2.70	2.81	1.67	1.14	1.68	1.00	0.84
PACA_AU_seq	9	12.79	7.79	3.40	2.93	1.45	2.17	1.13	0.93	1.76	0.74
Puleo_array	10	9.54	9.05	6.17	3.32	2.68	1.98	1.43	1.44	0.91	1.40

Table 15: K_effective per dataset and impact on hold-out C-index. K_eff fit uses point_normal with K = K_effective.

Dataset	K_eff	C Train K=5	C Test K=5	C Train K_eff	C Test K_eff	ΔC Test
CPTAC	8	0.315	0.389	0.315	0.389	0.000
Dijk	10	0.383	0.381	0.680	0.544	0.163
Moffitt_GEO_array	10	0.429	0.475	0.680	0.530	0.055
PACA_AU_array	8	0.353	0.405	0.677	0.738	0.333
PACA_AU_seq	9	0.210	0.185	0.840	0.741	0.555
Puleo_array	10	0.340	0.396	0.699	0.674	0.278
TCGA_PAAD	9	0.385	0.341	0.622	0.625	0.284

Interpretation. K_{eff} ranges from 8 to 10 across all 7 cohorts — every dataset benefits from more than the fixed 5 factors. Notably, three cohorts (Dijk, Moffitt, Puleo) saturate at $K_{\text{eff}} = 10 = K_{\text{max}}$, meaning the data supports even more latent structure than the 10-factor diagnostic fit can expose. For these cohorts, re-running with $K_{\text{max}} = 15$ on Longleaf HPC would be informative.

Factors above $K = 5$ cleared the activity threshold either because they explain non-trivial variance (PVE > 1%) or carry survival signal that was missed when the model was constrained to 5 dimensions. The positive ΔC_{test} values in the Unified Comparison table above confirm that recovering these additional factors improves out-of-sample survival prediction.

> **Note**
>
> **On cross-validation for K selection.** The auto-prune heuristic is practical (~8 s per cohort: one K_{max} fit + one K_{eff} refit) but relies on the threshold criterion ($|\hat{\beta}| > 0.05$ or PVE $> 1\%$), which is not guaranteed to minimise prediction error. A proper CV grid search over $K \in \{2, \dots, 10\}$ with 5 folds requires $8 \times 5 = 40$ fits per cohort. At ~20 s per fit, that is roughly **13 minutes per cohort** and **90 minutes total** — feasible on Longleaf HPC. The interface is stubbed as `select_K_cv()` in `code/select_K.R` (`K_SELECT=cv` in the runner); it errors on laptops with an informative message.

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14 Summary and Future Directions

14.1 What Was Done and Why It Matters

Pancreatic ductal adenocarcinoma has a five-year survival rate below 12%, in part because patients are not yet stratified at diagnosis into groups that would respond differently to available treatments. Reliable molecular prognostic signatures — gene programs whose activation levels predict survival — could guide treatment decisions and define patient subgroups for clinical trial enrollment. The challenge is that most molecular subtyping methods discover expression structure first (e.g., by PCA or clustering) and then test for survival association post hoc. This two-step approach is statistically conservative: it may miss programs whose variation is modest in expression space but strongly predictive of survival.

The **Supervised Bayesian Matrix Factorization (SBMF)** model addresses this by jointly decomposing the expression matrix and fitting the Cox proportional hazards model in a single variational inference framework. The K latent factors are simultaneously constrained to explain expression variance *and* maximise survival signal, producing gene expression programs (GEPs) that are both molecularly coherent and clinically relevant. Inference is exact in the variational sense: each update is guaranteed to increase the evidence lower bound.

This report applies SBMF to **seven PDAC cohorts spanning three assay platforms** — RNA-seq (TCGA_PAAD, Dijk, PACA_AU_seq), microarray (Moffitt, PACA_AU_array, Puleo_array), and proteomics (CPTAC) — demonstrating that the algorithm generalises across fundamentally different data types. Beyond the baseline fit, three methodological questions were systematically evaluated: (1) which sparsity prior best recovers prognostic structure in real genomics data; (2) how many latent factors each cohort actually supports; and (3) whether the learned factor–survival associations hold up on unseen patients.

14.2 Key Findings

Baseline results. All 7 cohorts converge successfully (14–259 iterations). The largest single-dataset signal comes from Puleo_array (n=288): three factors with significant survival stratification. CPTAC identifies a dominant protective proteomic program ($\hat{\beta}_3 = -0.435$, log-rank p < 0.001). PACA_AU_seq, despite being the smallest RNA-seq cohort (n=52), achieves the highest in-sample C-index (0.788) owing to its nearly complete event follow-up (60% event rate).

Prior family. `point_laplace` at $K = 5$ outperforms the default `point_normal` prior on hold-out C-index in all 7 cohorts ($\Delta C_{\text{test}} = +0.06$ to $+0.70$). However, when both priors are compared at K_{eff} (the complete factorial), the advantage reverses for 4/7 cohorts — `point_normal` wins at K_{eff} in TCGA_PAAD, CPTAC, Dijk, and Puleo_array, while `point_laplace` wins at K_{eff} in Moffitt and PACA_AU_seq. The dominant driver of improvement across all conditions is K selection (using K_{eff} vs. $K = 5$), rather than prior family choice.

K selection. Fixing $K = 5$ was insufficient for every cohort: auto-prune with $K_{\text{max}} = 10$ identifies $K_{\text{eff}} \in \{8, 9, 10\}$ across all datasets. Three cohorts saturate at $K_{\text{eff}} = 10$, indicating the data supports even more latent factors. Refitting with K_{eff} consistently improves hold-out C-index over the $K = 5$ baseline.

Hold-out generalisation. Stratified 80/20 hold-out evaluation shows positive generalisation relative to the PCA baseline in 5/7 cohorts (clearest: PACA_AU_array, PACA_AU_seq, CPTAC, Puleo_array). PACA_AU_seq and TCGA_PAAD show the largest absolute improvements once the correct prior and K are used. The weakest generalisation is in cohorts with small test sets ($n_{\text{test}} \leq 15$), where C-index estimates are inherently noisy.

14.3 Current Limitations

1. **K saturation.** Three cohorts saturate at $K_{\text{eff}} = K_{\text{max}} = 10$. The true dimensionality of these datasets is unknown; a larger diagnostic fit ($K_{\text{max}} = 15$ or 20) on Longleaf HPC would establish a reliable ceiling.
2. **Hold-out set size.** With 20% test fractions, n_{test} ranges from 10 to 57. C-index estimates at these sample sizes have wide confidence intervals; observed improvements may not be statistically significant at the individual-cohort level.
3. **Pseudo-inverse projection.** Test-patient loadings $\hat{L}_{\text{test}}$ are estimated by least-squares projection (no EBNM step), which does not propagate posterior uncertainty from the training fit. A variational projection would give calibrated risk scores.
4. **Feature selection leakage.** The top-5,000 genes were selected by variance computed on the full dataset before splitting. A strict evaluation would apply variance filtering on training patients only; the current approach slightly inflates test performance.
5. **No pathway annotation.** The top-feature lists and GEP heatmaps identify candidate genes but do not yet assign biological pathway labels. Without enrichment analysis, it is not possible to say whether the prognostic programs correspond to known PDAC biology (e.g., Moffitt basal/classical subtypes, immune infiltration, stromal remodelling).
6. **ELBO not used for model selection.** The full variational ELBO (genomics likelihood + Cox likelihood + KL divergences) is now tracked at each iteration, but has not yet been used as the primary criterion for selecting K or comparing prior families. ELBO-based model comparison would provide a principled, sample-size-independent ranking of configurations.

14.4 Future Directions

1. **K selection via cross-validation.** Implement `select_K_cv()` on Longleaf HPC (~90 min total) to replace the threshold heuristic with a proper held-out selection criterion.
2. **Gene set enrichment analysis.** Submit the top-loaded genes per factor to fgsea or Enrichr against MSigDB, KEGG, and Reactome to assign biological pathway labels and connect the statistical GEPs to known PDAC biology.
3. **ELBO-based model selection.** The full variational ELBO (now tracked at every iteration via `history$elbo_full`) enables principled comparison across K, prior families, and datasets. The next step is to use ELBO as a model selection criterion — e.g., select K as the value that maximises held-out ELBO rather than the PVE/beta threshold heuristic.
4. **Vertical multi-modal integration.** CPTAC has paired RNA-seq and proteomics for the same patients. A shared-L model — where both Y_{RNA} and Y_{protein} are explained by the same loading matrix — would extract programs concordant at both the transcriptomic and proteomic level, with the survival model trained jointly on both modalities.
5. **Larger K and Longleaf runs.** Re-run saturating cohorts (Dijk, Moffitt, Puleo) with $K_{\text{max}} \geq 15$ to probe the true dimensionality of their prognostic expression space.